Topology And Its Applications

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Table of Contents

Lecture 1. Topology: A Quick Overview
Lecture 2. A Quick Overview (Cont.)
Lecture 3. A Quick Overview (End)
Lecture 4. Advanced Metric Space Topics
Lecture 5. Advanced Metric Space Topics (Cont.)
Lecture 6. The Banach Contraction Theorem
Lecture 7. Metrics for Spaces of Maps
Lecture 8. Applications of the Contraction Principle
Lecture 9. Applications of the Contraction Principle (Cont.)
Lecture 10. Topological and Normed Vector Spaces
Lecture 11. Topological and Normed Vector Spaces (Cont.)
Lecture 12. Compactness for Metric Spaces
Lecture 13. Compactness for Metric Spaces. (Cont.)
Lecture 14. Compactness for Metric Spaces, The End.
Lecture 15. The Field of Reals and its Compact Subsets
Lecture 16. Product Metric Spaces and The Ascoli-Arzela Theorem
Lecture 17. Topological Spaces
Lecture 18. The Urysohn and Tietze Extension Theorems
Lecture 19. Urysohn and Tietze Theorems (Continued)
Lecture 20. Moore-Smith Convergence
Lecture 21. Moore-Smith Convergence (Cont.)
Lecture 22. Product Spaces and Tychononoff's's Theorem
Lecture 23. Connected Spaces and Sets
Lecture 24. Connected Spaces and Sets (Cont.)
Lecture 25. Homotopy Theory
Lecture 26. Fiber Bundles and Covering Spaces
Lecture 27. Liftings and the Covering Homotopy Theorem
Lecture 28. Applications of The Covering Homotopy Theorem
Lecture 29. Applications of The Non-triviality of $\Pi_1(S^1; 1)$

Lecture 1.
Topology: A Quick Overview

1.1. What is Topology All About?

In many parts of Mathematics, it is useful to have some concept of "closeness" relating two points of a set, and in a very broad sense, *Topology*, or more precisely *Point Set Topology*, is the study of how to implement such a concept and how to make effective use of it. In particular, once one has a concept of closeness for the points of two sets X and Y, there is a very important associated concept of "continuity" for functions $f : X \to Y$. Namely, f is said to be continuous at $x_0 \in X$ if $f(x)$ can be made arbitrarily close to $f(x_0)$ by requiring x to be sufficiently close to x_0, and f is called continuous if it is continuous at every point of X. (This may appear to be an imprecise, intuitive approximation to a more precise definition, but we will see that if we are careful about how we define our terminology then it is a mathematically precise definition.)

Our first task will be to find a good, general definition of what we mean by a *topological space*—or just a *space* for short. (Once we have done so, it will be obvious that we get a Category if we take the objects to be topological spaces and the morphisms to be the continuous maps between them.) As with most other mathematical categories, to make a set X into a topological space involves defining some sort of extra "structure" for the space that satisfies certain axioms, just as making a set G into a group involves specifying a "group law" for G.

$\triangleright$ **Exercise 1–1.** If you have not had an introduction to Categories, look here:

http://en.wikipedia.org/wiki/Category_(mathematics)#Definition

The definition of the additional structure $\mathcal{T}$ (called a *topology*) that makes a set X into a topological space is not at all complicated, but without some discussion it might look unmotivated, so we will provide a little background before defining it below.

The first topological space most people encounter is the real line $\mathbf{R}$. The distance between two real numbers x and y is defined to be

$\rho(x,y) := |x-y|$, and the two points are considered "close" if the distance between them is small. Similarly, the distance between two points $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$ in $\mathbf{R}^n$ is defined to be their usual Euclidean distance $\rho(x,y) := ||x-y||$ (where of course the "norm" function $||\ ||$ is defined by $||v||^2 := v_1^2 + \dots v_n^2$) and again x and y are considered "close" if that distance is small.

So it begins to look like the extra piece of structure we need to associate with a set X in order to define a notion of "being close" is a distance or metric function ρ_X from $X \times X$ into $\mathbf{R}$, and in fact for many years that was how topological notions were handled. Let us recall the conditions that such a function ρ is assumed to satisfy in order to qualify as an acceptable "metric":

1) Symmetry: $\rho_X(x,y) = \rho_X(y,x)$
2) Positivity: $\rho_X(x,y) \geq 0$ and $\rho_X(x,y) = 0$ if and only if $x = y$.
3) Triangle Inequality: $\rho_X(x,z) \leq \rho_X(x,y) + \rho_X(y,z)$.

The first two of these conditions are pretty obvious things to demand of anything one would intuitively want to call a "distance"—they just say that for example the distance from New York to Boston is the same as the distance from Boston to New York, and that all distances are non-negative numbers and are positive for distinct points. But what about the third condition? Note that its intuitive content is clear. I think of it as saying that "Things (x and y) close to the same thing (z) are close to each other." As we shall see, it is the key property that makes the definition of a metric work, and in any case we define a *Metric Space* to be a set X together with a function $\rho_X : X \times X \to \mathbf{R}$ satisfying the above three conditions for all $x, y, z \in X$. (Normally, when there is no chance of confusion, we will drop the subscript on ρ_X.)

Note that our condition for a function $f : X \to Y$ between metric spaces to be continuous at $x_0 \in X$ becomes, "$\rho_Y(f(x), f(x_0))$ can be made arbitrarily small by requiring that $\rho_X(x, x_0)$ is sufficiently small", or in First Year Calculus Speak: For any $\epsilon > 0$, there exists a $\delta > 0$ such that $\rho_Y(f(x), f(x_0)) < \epsilon$ provided $\rho_X(x, x_0) < \delta$.

1.2. Open Sets and Neighborhoods in a Metric Space.

Next we recall some standard definitions from metric space theory.

Definition. If X is a metric space, $x_0 \in X$, and $\epsilon > 0$, we define $B_\epsilon(x_0, X)$, *the* ϵ*-ball about* x_0 *in* X, to be the set of all $x \in X$ satisfying $\rho(x, x_0) < \epsilon$. (We will often just write $B_\epsilon(x_0)$ if X is clear from the context.) A subset O of a metric space X is called an *open subset of* X if whenever a point x_0 belongs to O it follows that some ϵ-ball about x_0 is included in O

Terminology. We will say that something is true for all x in a metric space that are *sufficiently close to* x_0 if it is true for all x in some $B_\epsilon(x_0, X)$.

Using this terminology, we can now say that a subset O of a metric space X is an open subset of X if whenever a point x_0 belongs to O it follows that all points sufficiently close to x_0 also belong to O.

Definition. If X is a metric space and $x_0 \in X$ then a subset N of X is called a neighborhood of x if all points sufficiently close to x_0 belong to N.

So, a subset of a metric space is open if and only if it is a neighborhood of each of its points.

▷ **Exercise 1–2.** Show that if X is a metric space then:

a) the empty set, $\emptyset$, and the whole space, X, are both open sets of X.

b) the intersection of two open sets is open, and hence, by induction, any finite intersection of open sets is open. (Here is where you finally get to use the triangle inequality.)

c) the union of *any* collection of open subsets of X is open.

Remark. Later we will generalize the above exercise to define what we mean by a "topological space": it will be a set X together with a collection of subsets $\mathcal{T}$, called the open sets of X, that satisfy a), b), and c). For this reason, where possible, we will attempt to use only these properties to frame further definitions and prove further facts about metric spaces. This will insure that such results generalize to topological spaces. A metric space concept that can be formulated entirely in terms of open sets, without explicit reference to metrics is said to be *topological*, so for example the next two exercises shows that *neighborhood* and *continuity* are topological concepts.

▷ **Exercise 1–3.** Show that N is a neighborhood of x in X if and only if there is an open set O of X with $x \in O \subseteq N$.

▷ **Exercise 1–4.** Let X and Y be metric spaces and $f : X \to Y$ a function between them.

1) Show that a necessary and sufficient condition for f to be continuous at $x_0 \in X$ is that if N is any neighborhood of $f(x_0)$ it follows that $f^{-1}(N)$ is a neighborhood of x_0. (Recall that for $A \subseteq Y$, $f^{-1}(A)$, the inverse image of A by f, is defined to be the subset of X consisting of all x such that $f(x) \in A$.)
2) Deduce that f is continuous if and only if the inverse image $f^{-1}(O)$ of every open set O of Y is an open set of X.

▷ **Exercise 1–5.** Show that two distinct points in a metric space have disjoint neighborhoods. (A topological space with this property is called a *Hausdorff space*).

Since the union of open sets of a space X is open, if A is any subset of X there is a *largest* open set included in A, namely the union of *all* open subsets of A. This set is called the *interior* of A and is usually denoted either by A° or $\mathrm{int}(A)$. Clearly a point p of A belongs to A° if and only if all points sufficiently close to p belong to A, i.e., if and only if A includes a neighborhood of p, and such a point p is called an *interior point* of A. It is also clear that a set is open if and only if it is equal to its interior, and hence $A^{\circ\circ} = A^\circ$.

Lecture 2.
A Quick Overview (Cont.)

2.1. Closed Sets in a Space.

If X is any set and A is a subset of X, then we denote the complement of A in X by $X \setminus A$ or $\complement_X(A)$. When X is clear from the context then we will write simply $\complement A$ for the complement of A. We note that $A \mapsto \complement A$ is a one-to-one map of the collection of all subsets of X onto itself—in fact it is its own inverse. Moreover it reverses inclusions, that is, if $A \subseteq B$ then $\complement B \subseteq \complement A$.

Definition. A subset F of a space X is called a *closed* subset of X if $\complement F$, its complement in X, is an open subset of X

Remarks. As someone put it, unlike doors—which are always either open or closed and never both, the subsets of a space can be open, closed, neither, or both! Recall in particular that the empty set and the whole space are always open, and since they are complements of each other, they are also both closed. (As we shall see later, a space is called *connected* precisely when these are the *only* open and closed sets.)

Next, assume that X and Y are sets and that f is a function from X into Y. It is easy to check that for any $A \subseteq Y$, $\complement_X f^{-1}(A) = f^{-1}(\complement_Y A)$. (This merely says that, for any point x of X, $f(x)$ is not in A if and only if $f(x)$ is in the complement of A !) Since $\complement_Y$ interchanges the open and closed subsets of Y and similarly $\complement_X$ interchanges the open and closed subsets of X it follows that if $f^{-1}(A)$ is open in X whenever A is open in Y, then $f^{-1}(A)$ is closed in X whenever A is closed in Y. It now follows from an exercise above that:

2.1.1 Proposition. *If X and Y are spaces, a function $f : X \to Y$ is continuous if and only if the inverse image of every closed set of Y is a closed set of X.*

In Logic there is an (obvious) principle called "De Morgan's Laws" stating that if P and Q are two statements, then NOT(P AND Q) is equivalent to (NOT P) OR (NOT Q) and similarly, NOT(P OR Q) is equivalent to (NOT P) AND (NOT Q). If we apply this to the two statements "$x \in A$" and "$x \in B$", it says that "the complement of the intersection is the union of the complements" and that dually, "the complement of the union is the intersection of the complements", or in symbols: $\complement(A \cap B) = (\complement A \cup \complement B)$ and $\complement(A \cup B) = (\complement A \cap \complement B)$. It is easy to see that the same holds for infinite intersections and unions, so one can simply think of De Morgan's Law as saying that "complementation turns union into intersection and vice versa". Keeping this in mind, the fact that finite intersections and arbitrary unions of open sets are open is equivalent to the following:

2.1.2 Proposition. *Finite unions and arbitrary intersections of closed sets are again closed sets.*

So, if S is a subset of a space X, then just as there is a largest open set included in S (its interior) obtained by taking the union of all the open sets included in S, similarly there is a smallest closed set that includes S, obtained by taking the intersection of all the closed sets that include S.

Definition. If S is a subset of a space X, then the *closure* of S (in X), denoted by $\bar{S}$, is the smallest closed subset of X that includes S.

Clearly S is closed if and only if $S = \bar{S}$, so in particular $\bar{\bar{S}} = \bar{S}$.

Definition. If S is a subset of a space X, then a point p of X is said to be *adherent* to S if there are points of S arbitrarily close to p, i.e., if every neighborhood of p meets S.

If $p \notin \bar{S}$, then $\complement\bar{S}$ is an open set containing p, and hence it is a neighborhood of p that does not meet S, so p is not adherent to S. On the other hand, if $p \in \bar{S}$ and N is a neighborhood of p, then $O = N^\circ$ is an open neighborhood of p, hence it must meet S (since otherwise $\complement O$

would be a closed set that includes S–and hence includes $\bar{S}$— but does not contain p, a contradiction), so p is adherent to S. This proves:

2.1.3 Proposition. *If S is a subset of a space X then $\bar{S}$, its closure in X, consists of all points of X that are adherent to S.*

Lecture 3.
A Quick Overview (End)

Definition. If A is a subset of a space X then a subset D of A is said to be *dense in* A if every point of A is adherent to D, or equivalently if A is included in the closure of D. A space X is said to be *separable* if there exists a countable set that is dense in X.

So, for example, $\mathbf{R}^n$ is separable, since the points with rational coordinates are clearly a countable dense subset.

If p is a point of a metric space X and S is a subset of X, then we define the distance from S to p to be the greatest lower bound of the distances $\rho(p, s)$ from p to a point $s \in S$. More generally, if A and B are subsets of a metric space X, then the distance from A to B is the greatest lower bound of the distances $\rho(a, b)$ for $a \in A$ and $b \in B$.

▷ **Exercise 3–1.**

1) Show that if S is a subset of a metric space X, then its closure $\bar{S}$ consists of all the points of X having distance zero from S.
2) If A and B are closed sets and their distance apart is zero, does it follow that A and B have a common point? Either prove this is always the case or find a counter-example.

Remark. If X is a space and A is any subset of X, then clearly $S \mapsto \complement S$ is a one-to-one map of all the open subsets of X included in A with all the closed subsets of X that include $\complement A$.

▷ **Exercise 3–2.** Show that if A is a subset of a space X then the complement of its interior is the closure of its complement. That is $\complement A^{\circ} = \overline{\complement A}$. (Hint: Use the above remark.) Show by a simple formal manipulation that this implies the complement the closure of A is the interior of its complement, or $\complement \bar{A} = (\complement A)^{\circ}$.

Remark. There is a name for the complement of the closure of A—it is called the *exterior* of A. (This is reasonable, since by the above exercise it is the interior of the complement of A.) The set $\bar{A} \cap \overline{\complement A}$, consisting of points that are adherent both to A and its complement, also has a natural name, the *boundary* of A (also sometimes called the *frontier* of A). It is denoted by ∂A.

▷ **Exercise 3–3.** If X is a metric space and $x_0 \in X$, then show that the function $f : X \to \mathbf{R}$ defined by $f(x) := \rho(x, x_0)$ is continuous.

Definition. If X is a metric space, $x_0 \in X$ and $r \geq 0$, then we define $\bar{B}_r(x_0, X) := \{x \in X \mid \rho(x, x_0) \leq r\}$, and call it *the closed ball of radius r at x_0*.

It is natural to conjecture that $\bar{B}_r(x_0, X)$ is indeed a closed set, and even that it is equal to the closure $\overline{B_r(x_0, X)}$ of the corresponding open ball, but:

▷ **Exercise 3–4.**

1) Use the preceding exercise to conclude that $\bar{B}_r(x_0, X)$, is always a closed subset of X that includes $\overline{B_r(x_0, X)}$.
2) Show by example that $\bar{B}_r(x_0, X)$ and $\overline{B_r(x_0, X)}$ need *not* be equal.

3.1. Sequences in a Metric Space.

Terminology. We will say that the elements of a sequence $\{x_n\}$ have a certain property *eventually* if all the x_n for n sufficiently large have the property, and we say that the elements of the sequence satisfies a property *frequently* if there are arbitrarily large values of n for which x_n has the property.

▷ **Exercise 3–5.** Show that if the elements of a sequence eventually satisfy a property P and also eventually satisfy a property Q, then they also eventually satisfy the property P AND Q.

You may recall that a sequence $\{x_n\}$ in $\mathbf{R}$—or more generally in a metric space X—is said to converge to a point x_0 (written $\lim_{n\to\infty} x_n = x_0$) if given any $\epsilon > 0$, there exists a positive integer N such that $n > N$ implies $\rho(x_n, x_0) < \epsilon$. Using the above terminological conventions we can rewrite this in a much more intuitive and easy to remember form, namely, $\lim_{n\to\infty} x_n = x_0$ if and only if, for any neighborhood N of x_0, the x_n are eventually in N. And writing the definition of a limit in this form has another advantage: since it does not refer explicitly to a metric, but only to the topological concept of a neighborhood of a point, it makes sense for any topological space.

Let's look at the classic proof that limits of sequences are unique, but using our above terminological conventions to make the proof look more intuitive. We suppose that a sequence converges to both p_1 and p_2 where $p_1 \neq p_2$, and we choose disjoint neighborhoods N_1 and N_2 of p_1 and p_2 respectively. Since the sequence converges to p_1 it is eventually in N_1, and since it converges to p_2 it is also eventually in N_2, hence by the previous exercise it is eventually both in N_1 and in N_2, i.e., in their intersection, contradicting that N_1 and N_2 are disjoint.

3.1.1 Proposition. *If A is a subset of a metric space X, then a necessary and sufficient condition for a point x_0 of X to be adherent to A is that some sequence of points of A converge to x_0.*

Proof. If $\{a_n\}$ is a sequence of points of A converging to x_0 and N is any neighborhood of x_0, then eventually the a_n are in N, so x_0 is adherent to A.

Conversely, if x_0 is adherent to A, then for each positive integer n, $B_{\frac{1}{n}}(x_0, X)$ is a neighborhood of x_0 and hence intersects A in some point, say a_n, and since $\rho(a_n, x_0) < \frac{1}{n}$ it follows that a_n converges to x_0. ∎

3.1.2 Corollary. *If A is a subset of a metric space X, then a subset D of S is dense in S if and only if every point of S is a limit of a sequence of points of D.*

3.1.3 Proposition. *Let X and Y be metric spaces and $f : X \to Y$ a function between them. A necessary and sufficient condition for f to be continuous at $x_0 \in X$ is that whenever a sequence $\{x_n\}$ converges to x_0 it follows that the image sequence $\{f(x_n)\}$ converges to $f(x_0)$.*

Proof. Assume f is continuous at x_0 and that $\{x_n\}$ converges to x_0. If N is any neighborhood of $f(x_0)$, then by continuity, $f^{-1}(N)$ is a neighborhood of x_0, so x_n is eventually in $f^{-1}(N)$, i.e., $f(x_n)$ is eventually in N, so $\{f(x_n)\}$ converges to $f(x_0)$.

Conversely, if f is not continuous at x_0 then there is a neighborhood N of $f(x_0)$ such that $f^{-1}(N)$ is not a neighborhood of x_0, so in particular $B_{\frac{1}{n}}(x_0, X)$ is not included in $f^{-1}(N)$, i.e., there exists x_n with $\rho_X(x_n, x_0) < \frac{1}{n}$ and $f(x_n) \notin N$. Thus $\{x_n\}$ converges to x_0, but $\{f(x_n)\}$ does not converge to $f(x_0)$. ∎

3.1.4 Corollary. *Let X and Y be metric spaces and let $f : X \to Y$ be a function between them. A necessary and sufficient condition for f to be continuous is that whenever a sequence $\{x_n\}$ in X converges to a point x_0, it follows that $\{f(x_n)\}$ converges in Y to $f(x_0)$..*

This completes our quick overview of the basic elements of point-set topology. We next treat some more advanced topics concerning metric spaces.

Lecture 4. Advanced Metric Space Topics

4.1. The First and Second Axioms of Countability

In the two propositions at the end of Lecture 3 the first part of the proof uses only the topological concept of neighborhood and works even for spaces that are not metric. However the second half uses the sequence $B_{\frac{1}{n}}(x_0, X)$ of metric balls to construct an appropriate sequence, and these propositions are in fact **not** valid for arbitrary non-metric spaces. However we discuss next a class of spaces more general than metric spaces for which we can construct a sequence of neighborhoods of any point x_0 analogous to the $B_{\frac{1}{n}}(x_0, X)$ and use it to generalize the second half of the above proofs for these spaces.

The First Axiom of Countability.

Definition. A collection C of neighborhoods of a point p in a space X is said to be a *neighborhood basis* for p if every neighborhood of p includes an element of C, and a space X is said to satisfy the *First Axiom of Countability* if every point of X has a countable neighborhood basis.

For example, for a metric space X, the $B_\epsilon(x_0, X)$ form a neighborhood basis for p, and the $B_{\frac{1}{n}}(x_0, X)$ are a countable neighborhood basis (so a metric space satisfies the First Axiom of Countability).

If X satisfies the First Axiom of Countability and $p \in X$ then we can find a sequence N_k of neighborhoods of p such that every neighborhood N of p includes some N_k. Moreover, by replacing N_k by the intersection of $N_1, N_2, \ldots, N_k$, we can assume that the N_k are *nested*, i.e., that $N_{k+1} \subseteq N_k$, and then we call the N_k a nested neighborhood basis for p. If the N_k form a nested neighborhood basis for p and N is any neighborhood of p, then eventually $N_k \subseteq N$.

Notice that in a metric space the sequence of balls $N_k = B_{\frac{1}{k}}(p, X)$ is a nested neighborhood basis for p. Moreover we can use any nested neighborhood basis for x_0 to replace the use of $B_{\frac{1}{n}}(x_0, X)$ in the two propositions at the end of Lecture 3 .

▷ **Exercise 4–1.** Carry out the details of generalizing the two propositions at the end of Lecture 3 for spaces that satisfy the First Axiom of Countability.

The Second Axiom of Countability.

Definition. A collection $\mathcal{B}$ of open sets in a space X is said to be a *basis* for the open sets of X if every open set O of X is a union of a family of sets from $\mathcal{B}$, and a space X is said to satisfy the *Second Axiom of Countability* if there is a countable basis for the open sets of X. (A space satisfying the Second Axiom of Countability is often referred to as being a "second countable space".)

▷ **Exercise 4–2.** Show that if $\mathcal{B}$ is a basis for the open sets of X, then for each $p \in X$, the set C of open sets O of $\mathcal{B}$ that contain p is a neighborhood basis for p, and deduce that a space that satisfies the Second Axiom of Countability also satisfies the First Axiom of Countability.

▷ **Exercise 4–3.** Show that if $\mathcal{B}$ is a basis for the open sets of X and D is a set that contains at least one point in each set of $\mathcal{B}$, then D is dense in X, and deduce that a second countable space is separable (i.e., has a countable dense subset.)

▷ **Exercise 4–4.** Show that a metric space is second countable if and only if it is separable.

4.2. Cauchy Sequences and Completeness

In all that follows, X is a metric space with metric ρ.

Definition. A sequence $\{x_n\}$ in a metric space X is called a *Cauchy sequence* if $\rho(x_n, x_m)$ tends to zero as n and m both tend to infinity; that is if for any $\epsilon > 0$, $\rho(x_n, x_m) < \epsilon$ provided both n and m are sufficiently large.

$\triangleright$ **Exercise 4–5.** Show that a convergent sequence is Cauchy. (Hint: remember that "things close to the same thing are close to each other".)

$\triangleright$ **Exercise 4–6.** Show that if a Cauchy sequence has a convergent subsequence then it is convergent.

Definition. A metric space is called *complete* if every Cauchy sequence in X is convergent.

Remark. We recall that it follows easily from the Least Upper Bound Principle that the space $\mathbf{R}$ of real numbers (with the usual metric $\rho(x, y) = |x - y|$) is complete—and conversely, the Least Upper Bound Principle is an easy consequence of completeness. As we will see later when we discuss product spaces, the product of two complete spaces is complete, so it follows that $\mathbf{R}^n$ is complete for all n.

Completeness is an extremely important tool in Analysis where one often finds an object with some desirable property by constructing a sequence of objects that have better and better approximations to the given property, showing that the sequence is Cauchy, and taking its limit. We will see many examples of this as we proceed. And fortunately, the completeness or $\mathbf{R}$ allows us to demonstrate that many other important spaces that arise in Analysis are also complete.

$\triangleright$ **Exercise 4–7.** Show that a closed subspace of a complete space is complete, and show that a complete subspace Y of any space X is closed in X.

We recall that a sequence A_n of subsets of X is said to be *nested* if $A_{n+1} \subseteq A_n$ for all n, and that the *diameter*, $\operatorname{diam}(A)$, of a subset A of X is the least upper bound of the distances $\rho(a_1, a_2)$ for $a_1, a_2 \in A$.

The following result is frequently useful.

4.2.1 Ascoli's Lemma. *Let $\{F_n\}$ be a nested sequence of non-empty closed subsets of a complete metric space X. If $\operatorname{diam}(F_n) \to 0$ then the intersection of the F_n is a unique point x of X.*

Proof. Choose $x_n \in F_n$. By nestedness, both x_n and x_m belong to F_n if $m > n$, and since $\operatorname{diam}(F_n) \to 0$, it follows that $\{x_n\}$ is a Cauchy sequence. Since X is complete, $x_n \to x \in X$. Since each F_m is closed and the sequence is eventually in F_m, it follows that the limit x is also in F_m. The fact that the diameters of the F_n tend to zero clearly implies that their intersection has diameter zero, so it cannot contain more than one point. ∎

The following important result has many interesting consequences.

4.2.2 Baire Category Theorem. *If $\{O_n\}$ is a sequence of dense, open subsets of a complete metric space X, then their intersection $C = \bigcap_{n=1}^{\infty} O_n$ is also dense in X.*

Proof. We must show that any non-empty open subset U of X contains a point of C, i.e., a point that is in every O_n. We will construct a nested sequence F_n of non-empty closed sets such that $F_1 \subseteq U$, and $F_n \subseteq O_n$, and $\operatorname{diam}(F_n) \to 0$. Then, by the above lemma, the intersection of the F_n will be a point x that is clearly in U and also in every O_n.

We construct the F_n recursively. Since O_1 is dense and U is open, $U \cap O_1$ is non-empty, and since it is also open, it contains some closed ball, $F_1 = \bar{B}_{r_1}(x_1, X) \subseteq U \cap O_1$, where $0 < r_1 < 1$. Applying the same argument, but now with $B_{r_1}(x_1, X)$ replacing U, we can find a closed ball $F_2 = \bar{B}_{r_2}(x_2, X) \subseteq B_{r_1}(x_1, X) \cap O_2$, and with $0 < r_2 < \frac{1}{2}$, and inductively a closed set $F_n = \bar{B}_{r_n}(x_n, X) \subseteq B_{r_{n-1}}(x_{n-1}, X) \cap O_n$, and with $0 < r_n < \frac{1}{n}$ ∎

Note that a subset D of X is dense if and only if every open set meets D, i.e., if and only if $\complement D$ has no interior points. Thus, the complement of an open dense set is a closed set without interior, so taking complements of the O_n in the Baire Category Theorem gives the following equivalent dual version.

4.2.3 Corollary. *If $\{F_n\}$ is a sequence of closed sets with empty interiors in a complete metric space X, then their union $C = \bigcup_{n=1}^{\infty} F_n$ also has no interior points.*

Definition. A subset A of X is called *nowhere dense* if $\bar{A}$ has empty interior.

So another version of the above corollary is:

4.2.4 Corollary. *If $\{A_n\}$ is a sequence of nowhere dense subsets of a complete metric space X, then their union $C = \bigcup_{n=1}^{\infty} A_n$ also has no interior points.*

Remark. Note that in particular, each point in $\mathbf{R}$ is a closed set without interior, so it follows from the fact that $\mathbf{R}$ is complete that it must be uncountable !

Lecture 5.
Advanced Metric Space Topics (Cont.)

5.1. Lindelöf's Theorem

A collection of subsets of a set X whose union is all of X is called a *covering* (or sometimes just a *cover*) of X. The following result logically belongs a little earlier—in the section on the First and Second Axioms of Countability.

5.1.1 Lindelöf's Theorem. *If a space X is second countable then every covering of X by open sets has a countable sub-covering.*

Proof. Let $\mathcal{B} = \{O_j \mid j \in \mathbb{N}\}$ be a countable base for the open sets of X and let $\{U_\alpha \mid \alpha \in A\}$ be an open cover of X. Define J to be the set of $j \in \mathbb{N}$ such that O_j is included in U_α for some $\alpha \in A$, and for each $j \in J$ choose an $\alpha(j) \in A$ such that $O_j \subseteq U_{\alpha(j)}$. Then the $U_{\alpha(j)}$ are a countable subfamily of $\{U_\alpha \mid \alpha \in A\}$ and it it will suffice to show that they cover X, i.e., that if x is any point of X, then x belongs to some $U_{\alpha(j)}$. In any case, since the U_α cover X, x is in some U_α, and since the O_j are a basis for the open sets, this U_α is a union of certain O_j, so in particular for some j, $x \in O_j \subseteq U_\alpha$. Thus $j \in J$, so $x \in O_j \subseteq U_{\alpha(j)}$. ∎

5.2. Mappings Between Metric Spaces

If X and Y are topological spaces and $f : X \to Y$ a mapping between them, then about the only special property of f that we can define using the topologies of X and Y is continuity. However, if X and Y are metric spaces there are finer gradations of continuity.

Definition. If X and Y are metric spaces, $f : X \to Y$ is said to be:

a) *uniformly continuous* if given any $\epsilon > 0$ there exists a $\delta > 0$ such that if $x_1, x_2 \in X$ and $\rho_X(x_1, x_2) < \delta$, then $\rho_Y(f(x_1), f(x_2)) < \epsilon$.

b) *Lipschitz* if $\rho_Y(f(x_1), f(x_2)) \leq K\rho_X(x_1, x_2)$ for some $K > 0$ and all $x_1, x_2 \in X$. Such a K is called a *Lipschitz constant* for f and the above inequality is called a *Lipschitz Condition*.

▷ **Exercise 5–1.** Show that:
Lipschitz $\implies$ uniformly continuous $\implies$ continuous.

Definition. If X and Y are metric spaces, $F : X \to Y$ is said to be a *contraction mapping* if it satisfies a Lipschitz condition with Lipschitz constant $K < 1$. Such a Lipschitz constant K is then also referred to as a *contraction constant* for f.

The reason for singling out contraction mappings is that there is a very pretty and useful fact, called the *Banach Contraction Theorem*, that applies to a contraction mapping F of a complete metric space X to itself. Namely any such F has a unique *fixed point* (i.e., a point $x_0 \in X$ such that $F(x_0) = x_0$), and moreover there is a very simple algorithm for locating x_0; just choose any $x \in X$ and take the limit of its sequence of "iterates", under F, that is of $F(x), F(F(x)), F(F(F(x))), \ldots$.

▷ **Exercise 5–2.** The cosine function clearly maps the interval $[-1, 1]$ into itself. Use the Intermediate Value Theorem to prove that it is a contraction. Put a hand calculator into radian mode and keep pressing the cos button until the value stays fixed to six decimal places.

The fixed point equation, $F(x_0) = x_0$, may seem very special, but as we shall see in the following lectures, when we prove the Contraction Theorem and study some of its applications, the solution to a great many highly interesting and important problems in analysis can be reduced to solving some related fixed point equation. The next exercise gives an example of this and is meant to whet your appetite.

▷ **Exercise 5–3.** Suppose we want to find a continuous function $x(t)$ that satisfies the differential equation $\frac{dx}{dt} = f(x,t)$ with the initial condition $x(t_0) = x_0$. By integrating the differential equation from t_0 to t, see if you can (at least formally) convert this initial value problem into a fixed point equation for a function F from a space of continuous functions to itself.

If we want to make rigorous sense out of the above exercise, then we are going to have to define a metric for spaces of continuous functions in order to apply the Contraction Theorem. We will consider this after outlining the proof of the Contraction Theorem in a series of exercises.

Lecture 6.
The Banach Contraction Theorem

6.1. Exercises Leading to The Contraction Theorem

In what follows, X is a metric space and $f : X \to X$ is a contraction mapping with contraction constant K, i.e., $0 < K < 1$ and for all $x_1, x_2 \in X$,

$$\rho(f(x_1), f(x_2)) < K\rho(x_1, x_2).$$

So under the mapping f, the distance between any pair of distinct points gets contracted at least by a factor of K. Recall that the Banach Contraction Theorem says that if X is complete, then f has a unique fixed point x_0 and for any $x \in X$, the sequence of iterates of x under f converges to x_0. The following sequence of exercises will lead you through a proof of this theorem.

$\triangleright$ **Exercise 6–1.** Prove what I call the "Fundamental Inequality for Contraction Mappings", which says that if x_1 and x_2 are any two points of X, then

$$\rho(x_1, x_2) \leq \frac{1}{1-K}\Big(\rho(x_1, f(x_1)) + \rho(x_2, f(x_2))\Big).$$

Hint. This is **very** easy, and your proof should only be a few lines. Nevertheless, as we shall soon see, it is the key to the proof of the Banach Contraction Theorem.

Question. It is clear that the Fundamental Inequality requires that $K \neq 1$, but does it really require that $K < 1$?

Remark. The Fundamental Inequality says something a bit strange; namely that if we know how far f moves x_1 and how far it moves x_2, then from this knowledge alone we can estimate how far x_1 is from x_2.

$\triangleright$ **Exercise 6–2.** Prove that a contraction can have at most one fixed point.

Notation. For a positive integer k, $f^k : X \to X$ is f composed with itself k times. Using induction, $f^1(x) = f(x)$ and $f^{k+1}(x) = f(f^k(x))$.

Question. We also have $f^{k+1}(x) = f^k(f(x))$. Why is that?

▷ **Exercise 6–3.** What Lipschitz Condition does f^k satisfy? (More generally, if $f : X \to Y$ satisfies a Lipschitz condition with constant K and $g : Y \to Z$ satisfies a Lipschitz condition with constant K, what about $g \circ f : X \to Z$?)

▷ **Exercise 6–4.** Show that if X is any space (**not** necessarily metric), h is any continuous map of X to itself (**not** necessarily a contraction), and x is any point of X such that the sequence $\{h^n(x)\}$ of iterates converges, then the limit x_0 of this sequence, will be a fixed point of h. (Hint: The proof is a simple one-line argument.)

The final step we need to complete the proof of the Banach Contraction Theorem is contained in the following exercise.

▷ **Exercise 6–5.** Show that if x is any point of X then the sequence $\{f^n(x)\}$ of iterates is a Cauchy sequence.

Solution. Using the Fundamental Inequality for Contractions to estimate $\rho(f^n(x), f^m(x))$ gives:

$$\begin{aligned}
\rho(f^n(x), f^m(x)) \leq & \frac{1}{1-K}\Big(\rho(f(f^n(x)), f^n(x)) + \rho(f(f^m(x)), f^m(x))\Big) \\
\leq & \frac{K^n + K^m}{1-K}\rho(f(x), x)
\end{aligned}$$

so since $K < 1$, if both n and m go to infinity then $\rho(f^n(x), f^m(x)) \to 0$.

6.1.1 Banach Contraction Theorem. *If X is a complete metric space then any contraction mapping $f : X \to X$, has a unique fixed point p, and for any x in X the sequence $f^n(x)$ converges to p. In fact, if K is a contraction constant for f then $\rho(f^n(x), p) \leq \frac{K^n}{1-K}\rho(x, f(x))$.*

▷ **Exercise 6–6.** Prove this by putting together the results of the preceding exercises.

6.2. Practical Programming Considerations

If we can interpret the solution of an equation as the fixed point of a contraction mapping, then iteration is a simple algorithm for finding it. But something important is missing. Namely, when should we stop iterating and take the current value as the "answer". One possibility is to just keep iterating until the distance between two successive iterates is smaller than some predetermined "tolerance" (perhaps the machine precision). But this seems a little unsatisfactory, and there is actually a much neater "stopping rule".

Suppose we are willing to accept an "error" of ϵ in our solution, i.e., instead of the actual fixed point p of f we will be happy with any point p' of X satisfying $\rho(p, p') < \epsilon$. If we start our iteration at some point x in X, then it is easy to specify an integer N so that $p' = f^N(x)$ will be a satisfactory answer. The key, not surprisingly, lies in the Fundamental Inequality, which we apply now with $x_1 = f^N(x)$ and $x_2 = p$. It tells us that $\rho(f^N(x), p) \leq \frac{1}{1-K}\rho(f^N(x), f^N(f(x))) \leq \frac{K^N}{1-K}\rho(x, f(x))$, and since we want $\rho(f^N(x), p) \leq \epsilon$, we just have to pick N so large that $\frac{K^N}{1-K}\rho(x, f(x)) < \epsilon$. Now the quantity $d = \rho(x, f(x))$ is something that we can compute after the first iteration and we can then compute how large N has to be by taking the log of the above inequality and solving for N (remembering that $\log(K)$ is negative !). We can express our result as:

Stopping Rule. If $d = \rho(x, f(x))$ and

$$N > \frac{\log(\epsilon) + \log(1-K) - \log(d)}{\log(K)}$$

then $\rho(f^N(x), p) < \epsilon$.

From a practical programming point of view, this allows us to express our iterative algorithm with a "for loop" rather than a "while loop", but this inequality has another interesting interpretation. Suppose we take $\epsilon = 10^{-m}$ in our stopping rule inequality. What we see is that the growth of N with m is a constant plus $m/|\log(K)|$, or in other words, to get one more decimal digit of precision we have to do (roughly) $1/|\log(K)|$ more iteration steps. Stated a little differently, if we need N iterative steps to get m decimal digits of precision, then we need another N to double the precision to $2m$ digits.

Lecture 7. Metrics for Spaces of Maps

7.1. The ρ_∞ metric

In this section we again assume that X and Y are metric spaces, and we let Y^X denote the set of all functions $f : X \to Y$, and let $C(X, Y)$ denote the subset of continuous functions.

Definition. For $f_1, f_2 \in Y^X$ we define:

$$\rho_\infty(f_1, f_2) = \min(1, \sup\{\rho_Y(f_1(x), f_2(x)) \mid x \in X\}).$$

In words, $\rho_\infty(f_1, f_2)$ is the sup of the distances between images of f_1 and f_2 at corresponding points if that sup is less than 1, and otherwise it equals 1.

$\triangleright$ **Exercise 7–1.** Show that ρ_∞ is a valid metric for the set Y^X. (Hint: Recall that by Exercise 13, page 8 of our text, an exercise from the first homework assignment, $\bar{\rho}_Y(y_1, y_2) := \min(1, \rho_Y(y_1, y_2))$ is a valid metric for Y and is equivalent to ρ_Y.)

Definition. Let $\{f_n\}$ be a sequence in Y^X and let $f \in Y^X$. We say that f_n *converges pointwise to* f if for each $x \in X$ the sequence $f_n(x)$ converges in Y to $f(x)$; in other words if given $\epsilon > 0$ and $x \in X$, there exists a positive integer N (depending on ϵ and x) such that if $n > N$ then $\rho_Y(f_n(x), f(x)) < \epsilon$. On the other hand, we say that f_n *converges uniformly to* f if for every $\epsilon > 0$ there exists a positive integer N (**depending only on** ϵ) such that if $n > N$ then $\rho_Y(f_n(x), f(x)) < \epsilon$ for all $x \in X$.

Remark. Clearly uniform convergence implies pointwise convergence.

$\triangleright$ **Exercise 7–2.** Give an example of a sequence $f_n : \mathbf{R} \to \mathbf{R}$ that converges pointwise but NOT uniformly.

$\triangleright$ **Exercise 7–3.** Define what it means for a sequence $\{f_n\}$ in Y^X to be pointwise Cauchy and to be uniformly Cauchy, and show that uniformly Cauchy implies pointwise Cauchy.

$\triangleright$ **Exercise 7–4.** Show that if Y is complete then a pointwise Cauchy sequence $\{f_n\}$ in Y^X converges pointwise to a limit function $f \in Y^X$, and that if the sequence is uniformly Cauchy then the convergence of f_n to f is also uniform.

You have probably guessed what the above definition and exercises have to do with the metric ρ_∞.

$\triangleright$ **Exercise 7–5.** Show that a sequence $\{f_n\}$ in Y^X is uniformly Cauchy if and only if it is Cauchy with respect to the metric ρ_∞, i.e., if and only if $\lim_{m.n\to\infty} \rho_\infty(f_n, f_m) = 0$, and similarly this sequence converges uniformly to a function $f \in Y^X$ if and only $\lim_{n\to\infty} \rho_\infty(f_n, f) = 0$.

Then as a consequence of the two above exercises we have the following fact:

7.1.1 Proposition. *If Y is a complete metric space then so is Y^X with respect to the metric ρ_∞.*

7.1.2 Proposition. *The space $C(X, Y)$ of continuous maps of X into Y is a closed subspace of Y^X.*

$\triangleright$ **Exercise 7–6.** Prove the latter Proposition. (Note that what it says is that a uniform limit of a sequence of continuous functions is continuous. This is a fact that you should have seen in some degree of generality if you had a rigorous course in Analysis.)

$\triangleright$ **Exercise 7–7.** Give an example of a sequence of continuous functions $f_n : \mathbf{R} \to \mathbf{R}$ that converges pointwise to a discontinuous function.

7.1.3 Proposition. *If Y is a complete space then $C(X,Y)$ is also a complete metric space with respect to the metric ρ_∞.*

This is an easy corollary of the preceding proposition. Why? It is a fact that we will need when we apply the Banach Contraction Theorem in interesting ways.

Lecture 8. Applications of the Contraction Principle

8.1. Lipschitz Constants for Differentiable Maps

▷ **Exercise 8–1.** Use the Mean Value Theorem of differential calculus to show that if $X = [a, b]$ is a closed interval of $\mathbf{R}$ and $f : X \to \mathbf{R}$ is a continuously differentiable real-valued function on X then the maximum value of $|f'|$ is the smallest possible Lipschitz constant for f.

(So, for example, $\sin(1) \approx 0.841471 < 1$ is the best Lipschitz constant for the cosine function in $[-1, 1]$.)

Next suppose that V and W are orthogonal vector spaces, U is a convex open set in V, and $f : U \to W$ is a continuously differentiable map. Let's try to generalize the above exercise to find a Lipschitz constant for f. If p is in U then recall that Df_p, the differential of f at p, is a linear map of V to W defined by $Df_p(v) = (d/dt)_{t=0} f(p + tv)$, and it then follows that if $\sigma(t)$ is any smooth path in U then $d/dt f(\sigma(t)) = Df_{\sigma(t)}(\sigma'(t))$. If p and q are any two points of U, and $\sigma(t) = p + t(q - p)$ is the line joining them, then integrating from 0 to 1 gives:

8.1.1 The Finite Difference Formula.

$$f(q) - f(p) = \int_0^1 Df_{\sigma(t)}(q - p)\, dt.$$

Now recall that if T is any linear map of V to W, its norm $\|T\|$ is the smallest non-negative real number r such that $\|Tv\| \leq r\,\|v\|$ for all v in V. Since $\left\|\int_a^b g(t)\,dt\right\| \leq \int_a^b \|g(t)\|\,dt$, $\|f(q) - f(p)\| \leq (\int_0^1 \left\|Df_{\sigma(t)}\right\|\,dt)\,\|(q - p)\|$, and it follows that if $\left\|Df_{\sigma(t)}\right\| < K$ for $t \in [0, 1]$, then $\|f(q) - f(p)\| \leq K\,\|(q - p)\|$, so

8.1.2 Proposition. *Let V and W be orthogonal vector spaces, O open in V, and $f : O \to W$ a continuously differentiable map. If U*

is a bounded, convex, open set whose closure is included in O, then $\sup\{\|Df_p\| \mid p \in U\}$ *is a Lipschitz constant for f restricted to U.*

(The sup is finite because $\bar{U}$ is compact and $\|Df\|$ is continuous on $\bar{U}$.)

▷ **Exercise 8–2.** Show that this in fact gives the smallest Lipschitz constant for f in U.

8.2. The Inverse Function Theorem

Let V and W be orthogonal vector spaces and $g : V \to W$ a C^k map, $k > 0$. Suppose that for some v_0 in V the differential Dg_{v_0} of g at v_0 is a linear isomorphism of V with W. Then the Inverse Function Theorem says that g maps a neighborhood of v_0 in V one-to-one onto a neighborhood U of $g(v_0)$ in W, and that the inverse map from U into V is also C^k.

▷ **Exercise 8–3.** Show that it suffices to consider the case that v_0 and $g(v_0)$ are the respective origins of V and W, by replacing g by $v \mapsto g(v + v_0) - g(v_0)$. Then show that one can further reduce to the case that $W = V$ and Dg_0 is the identity mapping I of V by replacing this new g by $(Dg_0)^{-1} \circ g$.

Given y in V, define $f = f_y : V \to V$ by $f(v) = v - g(v) + y$. Note that a solution of the equation $g(x) = y$ is the same thing as a fixed point of f. We will show that if δ is sufficiently small, then f restricted to $X = N_\delta = \{v \in V \mid \|v\| \leq \delta\}$ is a contraction mapping of N_δ to itself provided $\|y\| < \delta/2$. By the Banach Contraction Principle it then follows that g maps N_δ one-to-one into V and that the image covers the neighborhood of the origin $U = \{v \in V \mid \|v\| < \delta/2\}$. This proves the Inverse Function Theorem except for the fact that the inverse mapping of U into V is C^k (which we will **not** prove).

The first thing to notice is that since $Dg_0 = I$, $(Df)_0 = 0$ and hence, by the continuity of Df, $\|(Df)_v\| < 1/2$ for v in N_δ provided δ is sufficiently small. Since N_δ is convex, by a remark above this proves that $\frac{1}{2}$ is a Lipschitz bound for f in N_δ, and in particular that f restricted to N_δ is a contraction. Thus it only remains to show that f maps N_δ

into itself provided $\|y\| < \delta/2$. That is, we must show that if $\|x\| \le \delta$ then also $\|f(x)\| \le \delta$. But since $f(0) = y$,

$$\begin{aligned}\|f(x)\| &\le \|f(x) - f(0)\| + \|f(0)\| \\ &\le \frac{1}{2}\|x\| + \|y\| \\ &\le \delta/2 + \delta/2 \le \delta. \qquad \blacksquare\end{aligned}$$

▷ **Exercise 8–4.** The first (and main) step in proving that the inverse function $h : U \to V$ is C^k is to prove that h is Lipschitz. That is, we want to find a $K > 0$ so that given y_1 and y_2 with $\|y_i\| < \delta/2$ and x_1 and x_2 with $\|x_i\| < \delta$, if $h(y_i) = x_i$ then $\|x_1 - x_2\| \le K\|y_1 - y_2\|$. Prove this with $K = 2$, using the fact that $h(y_i) = x_i$ is equivalent to $f_{y_i}(x_i) = x_i$ and that $\frac{1}{2}$ is a Lipschitz constant for $h = I - g$.

8.3. Ordinary Differential Equations (aka ODE)

What does the following sentence mean, and what image should it cause you to form in your mind?

> Let $V : \mathbf{R}^n \times \mathbf{R} \to \mathbf{R}^n$ be a time-dependent vector field, and let $x(t)$ be a solution of the differential equation $\frac{dx}{dt} = V(x,t)$ satisfying the initial condition $x(t_0) = x_0$.

Let us consider a seemingly very different question. Suppose you know the wind velocity at every point of space and at all instants of time. A puff of smoke drifts by, and at a certain moment you note the precise location of a particular smoke particle. Can you then predict where that particle will be at all future times?

We will see that when this somewhat vague question is translated appropriately into precise mathematical concepts, it leads to the above "differential equation", and that the answer to our prediction question translates to the central existence and uniqueness result in the theory of differential equations. (The answer, by the way, turns out to be a qualified "yes", with several important caveats.)

We interpret "space" to mean the n-dimensional real number space $\mathbf{R}^n$, so a "point of space" is just an n-tuple $x = (x_1, \ldots, x_n)$ of real numbers, and an "instant of time" will always be represented by a single real number t. Thus, knowing the wind velocity at every point of space and at all instants of time means that we have a function V that associates to each (x, t) in $\mathbf{R}^n \times \mathbf{R}$ a vector $V(x, t)$ in $\mathbf{R}^n$, the wind velocity at x at time t. We will denote the n components of $V(x, t)$ by $V_1(x, t), \ldots, V_n(x, t)$. (We will always assume that V is at least continuous, and usually that it is even continuously differentiable.)

How should we model the path taken by a smoke particle? An ideal smoke particle is characterized by the fact that it "goes with the flow", i.e., it is carried along by the wind. That means that if $x(t) = (x_1(t), \ldots, x_n(t))$ is its location at a time t, then its velocity at time t will be the wind velocity at that point and time, namely $V(x(t), t)$. But the velocity of the particle at time t is $x'(t) = (x_1'(t), \ldots, x_n'(t))$, where primes denote differentiation with respect to t, i.e., $x' = \frac{dx}{dt} = (\frac{dx_1}{dt}, \ldots, \frac{dx_n}{dt})$.

So the path of a smoke particle will be a differentiable curve $x(t)$ in $\mathbf{R}^n$ that satisfies the differential equation $x'(t) = V(x(t), t)$, or $\frac{dx}{dt} = V(x, t)$. If we write this in components it reads,

$$\frac{dx_i}{dt} = V_i(x_1(t), \ldots, x_n(t), t), \quad i = 1, \ldots, n$$

and for this reason it is often called a system of differential equations. Finally, if at a time t_0 we observe that the smoke particle is at the point x_0 in $\mathbf{R}^n$, then the "initial condition" $x(t_0) = x_0$ is also satisfied. The combination of a system of differential equations together with an initial condition is usually referred to as an "initial value problem" or IVP. In the next lecture we will see that with suitable conditions on the vector field V and on the interval of definition of the solution x, the Initial Value Problem for an ODE has a unique solution.

Lecture 9.
Applications of the Contraction Principle (Cont.)

9.1. Existence and Uniqueness for solutions of ODE

Let $V : \mathbf{R}^n \times \mathbf{R} \to \mathbf{R}^n$ be a C^1 time-dependent vector field on $\mathbf{R}^n$. In the following $I = [a, b]$ will denote a closed interval that contains t_0. We will denote by $C(I, \mathbf{R}^n)$ the vector space of continuous maps $\sigma : I \to \mathbf{R}^n$, made into a metric space with the distance function

$$\rho(\sigma_1, \sigma_2) = \max_{t \in I} \|\sigma_1(t) - \sigma_2(t)\| .$$

As we have seen, $C(I, \mathbf{R}^n)$ is a complete metric space. In fact, this just amounts to the theorem that a uniformly Cauchy sequence of continuous functions converges uniformly to a continuous function.

Define for each v_0 in $\mathbf{R}^n$ a map $F = F^{V,v_0} : C(I, \mathbf{R}^n) \to C(I, \mathbf{R}^n)$ by $F(\sigma)(t) := v_0 + \int_{t_0}^{t} V(\sigma(s), s)\, ds$. The Fundamental Theorem of Calculus gives $\frac{d}{dt}(F(\sigma)(t)) = V(\sigma(t), t)$, and clearly $F(\sigma)(t_0) = v_0$. It follows that if σ is a fixed point of F then it is a solution of the ODE $\sigma'(t) = V(\sigma(t), t)$ with initial condition v_0, and the converse is equally obvious. Thus it is natural to try to find a solution of this differential equation with initial condition v_0 by starting with the constant path $\sigma_0(t) \equiv v_0$ and applying successive approximations using the function F. We will now see that this idea works and leads to the following result, called the Local Existence And Uniqueness Theorem for C^1 ODE.

9.1.1 Theorem. *Let $V : \mathbf{R}^n \times \mathbf{R} \to \mathbf{R}^n$ be a C^1 time-dependent vector field on $\mathbf{R}^n$, $p \in \mathbf{R}^n$, and $t_0 \in \mathbf{R}$. There are positive constants ϵ and δ depending on V, p, and t_0 such that if $I = [t_0 - \delta, t_0 + \delta]$, then for each $v_0 \in \mathbf{R}^n$ with $\|v_0 - p\| < \epsilon$ the differential equation $\sigma'(t) = V(\sigma(t), t)$ has a unique solution $\sigma : I \to \mathbf{R}^n$with $\sigma(t_0) = v_0$.*

Proof. If $\epsilon > 0$, then using the technique explained earlier we can find a Lipschitz constant M for V restricted to the set of $(x, t) \in \mathbf{R}^n \times \mathbf{R}$

such that $\|x-p\| \leq 2\epsilon$ and $|t-t_0| \leq \epsilon$. Let B be the maximum value of $F(x,t)$ on this same set, and choose $\delta > 0$ so that $K = M\delta < 1$ and $B\delta < \epsilon$, and define X to be the set of σ in $C(I,V)$ such that $\|\sigma(t)-p\| \leq 2\epsilon$ for all $t \in I$. It is easy to see that X is closed in $C(I,V)$ and hence a complete metric space. The theorem will follow from the Banach Contraction Principle if we can show that for $\|v_0 - p\| < \epsilon$, F^{V,v_0} maps X into itself and has K as a Lipschitz bound.

If $\sigma \in X$ then for $t \in I$,

$$\|F(\sigma)(t) - p\| \leq \|v_0 - p\| + \int_{t_0}^{t} \|V(\sigma(s),s)\| \, ds \leq \epsilon + \delta B \leq 2\epsilon,$$

so F maps X to itself. And if $\sigma_1, \sigma_2 \in X$ then for $t \in I$,

$$\|V(\sigma_1(t),t) - V(\sigma_2(t),t)\| \leq M \|\sigma_1(t) - \sigma_2(t)\|, \text{ so}$$

$$\begin{aligned}
\|F(\sigma_1)(t) - F(\sigma_2)(t)\| &\leq \int_{t_0}^{t} \|V(\sigma_1(s),s) - V(\sigma_2(s),s)\| \, ds \\
&\leq \int_{t_0}^{t} M \|\sigma_1(s) - \sigma_2(s)\| \, ds \\
&\leq \int_{t_0}^{t} M\rho(\sigma_1,\sigma_2) \, ds \\
&\leq \delta M \rho(\sigma_1,\sigma_2) \leq K\rho(\sigma_1,\sigma_2),
\end{aligned}$$

and it follows that $\rho(F(\sigma_1), F(\sigma_2) \leq K\rho(\sigma_1,\sigma_2)$. ∎

▷ **Exercise 9–1.** A system of second order ODE is given by a function $F : \mathbf{R}^n \times \mathbf{R}^n \times \mathbf{R} \to \mathbf{R}^n$, and is usually written in the form

$$\frac{d^2 x_i}{dt^2} = F\Big(x_1, \ldots, x_n, \frac{dx_1}{dt}, \ldots, \frac{dx_n}{dt}, t\Big).$$

Typical examples of course come from Newton's Equations for various mechanical systems. The theory of systems of second order ODE can be subsumed under the theory of systems of first order ODE by regarding $(x,v,t) \mapsto (v, F(x,v,t))$ as a time-dependent vector field on $\mathbf{R}^{2n}$ and interpreting the above equation to mean the corresponding first order system:

$$\begin{aligned}\frac{dx_i}{dt} &= v_i \\ \frac{dv_i}{dt} &= F\Big(x_1, \dots, x_n, v_1, \dots, v_n, t\Big).\end{aligned}$$

With this in mind, what do you think a reasonable "initial value problem" for such a system of second order equations? Write the corresponding Existence and Uniqueness Theorem for this initial value problem.

Lecture 10. Topological and Normed Vector Spaces

10.1. Topological Vector Spaces

In what follows, "vector space" will always mean a vector space over either the real or complex field, but for simplicity we will assume that we are in the real case when stating definitions and theorems.

Definition. A topological vector space is a vector space V with a Hausdorff topology such that the addition map $V \times V \to V$ and scalar multiplication map $\mathbf{R} \times V \to V$ are continuous.

We will abbreviate topological vector space to TVS. The obvious examples are the familiar vector spaces $\mathbf{R}^n$, and it is an interesting and important fact (proved below) that these are the *only* finite dimensional examples. What this means is that if V is any n-dimensional vector space, then the linear isomorphism of V with $\mathbf{R}^n$ that you get by choosing a basis is automatically also a homeomorphism.

10.1.1 Lemma. *If V is a topological vector space and $v \in V$, then N is a neighborhood of 0 in V if and only if $N + v := \{u + v \mid u \in N\}$ is a neighborhood of v.*

▷**Exercise 10–1.** Prove this lemma. (Hint: What can you say about the translation map $\tau_v : V \to V$ defined by $\tau_v(u) := u + v$?)

When dealing with maps between topological vector spaces it is natural to emphasize linear maps because of the vector space structure, and continuous maps because of the topology, so the following fact simplifies many considerations.

10.1.2 Linear Continuity Principle. *Let $T : V \to W$ be a linear map between topological vector spaces. Then T is continuous at an* *arbitrary point $v \in V$, if and only if it is continuous at the origin, 0. Thus T is continuous if it is continuous at any one point.*

▷ **Exercise 10–2.** Prove this. (Hint: $T(v + u) - T(v) = T(u)$.)

10.1.3 Proposition. *Every linear map T of $\mathbf{R}^n$ into a topological vector space V is continuous.*

Proof. If $e_1, \ldots, e_n$ is the standard basis for $\mathbf{R}^n$ and $T(e_i) = v_i$, then $T(x_1, \ldots x_n) = \sum_{1=1}^{n} x_i v_i$, so the continuity of T follows directly from the continuity of addition and scalar multiplication in V. ∎

Definition. A non-empty subset S of a vector space V is called *balanced* if $s \in S$ and $|\alpha| \leq 1$ implies $\alpha s \in S$.

Remarks. A more geometric way of stating the definition is that S is balanced if for every point s of S, $-s$ is also in S and the line segment $[0, 1]s$ joining s to the origin is included in S. In particular the origin is in every balanced set. If S is a subset of a vector space V and A is a set of real numbers, then we will write $AS := \{as \mid a \in A \text{ and } s \in S\}$. With this notation the condition for S to be balanced is simply $[-1, 1]S \subseteq S$. It follows easily that if $X \subseteq V$ then for any $r > 0$, $(-r, r)X$ is balanced. Now if $s \neq 0$, then multiplication by s is clearly a homeomorphism of V. Hence if X is a neighborhood of 0, then since $(-r, r)X$ is just the union of the sets sX for $0 < s \leq r$, each of which is a neighborhood of 0, $(-r, r)X$ is a balanced neighborhood of zero.

Remark. The obvious examples of balanced sets are the balls $B_r(0, V)$ in any normed space V. In fact balanced neighborhoods of 0 play the role of such balls in TVS theory, so the following proposition is important.

10.1.4 Proposition. *If V is a topological vector space then every neighborhood of 0 in V includes a balanced neighborhood of 0.*

Proof. Let $M : \mathbf{R} \times V \to V$ denote the multiplication map. If U is any neighborhood of 0 in V, then by the continuity of M at the point

$(0,0)$, $M^{-1}(U)$ includes a set of the form $(-r,r) \times N$ where N is a neighborhood of 0 in V. But this just says that $(-r,r)N \subseteq U$, and as remarked above, $(-r,r)N \subseteq U$ is a balanced neighborhood of 0. ∎

10.1.5 Theorem. *Up to isomorphism there is only one n-dimensional topological vector space, and in fact, every linear isomorphism T of $\mathbf{R}^n$ with a topological vector space V is automatically bicontinuous. Consequently a linear map between two finite dimensional topological vector spaces is automatically continuous.*

Proof. We proved above that T is continuous, so by the Linear Continuity Principle it suffices to prove that T^{-1} is continuous at 0, and for that it will suffice to show that for any $r > 0$, $T(B_r(0, \mathbf{R}^n))$ is a neighborhood of 0 in V. Now $S_r := \{x \in \mathbf{R}^n \mid \|x\| = r\}$, the sphere of radius r in $\mathbf{R}^n$, is compact and therefor so is its image $T(S_r)$ in V, and since V is Hausdorff, $T(S_r)$ is closed in V, so its complement is a neighborhood of 0 in V, and by the proposition above there is a balanced neighborhood N of 0 in V included in the complement of $T(S_r)$. We claim that $N \subseteq T(B_r(0, \mathbf{R}^n))$, completing the proof. For if not then there exists a $v = T(x) \in N$ with $x \notin B_r(0, \mathbf{R}^n)$, i.e., $\|x\| \geq r$. Then since $s := \frac{r}{\|x\|} < 1$ and N is balanced, $sT(x) = T(sx) \in N$. But since $\|sx\| = r$, $T(sx) \in T(S_r)$ contradicting the fact that N is included in the complement of $T(S_r)$. ∎

10.1.6 Corollary. *Every finite dimensional subspace of a topological vector space is closed.*

Lecture 11. Topological and Normed Vector Spaces (Cont.)

11.1. Norms and Semi-norms

Definition. A real-valued function p on a vector space V is called a *semi-norm* for V if:

1) Positivity. $p(v) \geq 0$ for all $v \in V$.
2) Positive Homogeneity. $p(\alpha v) = |\alpha| p(v)$ for all $\alpha \in \mathbf{R}$ and $v \in V$.
3) Triangle Inequality. $p(v_1 + v_2) \leq p(v_1) + p(v_2)$ for all $v_1, v_1 \in V$.

   If in addition $p(v) = 0$ implies $v = 0$ then p is called a *norm* for V.

Remark. if p is a norm for V it is customary to write $\|v\|$, possibly with subscripts or primes to denote $p(v)$.

11.1.1 Lemma. *If p is a semi-norm then $|p(v_1) - p(v_2)| \leq p(v_1 - v_2)$.*

▷ **Exercise 11–1.** Prove this.

11.1.2 Proposition. *A semi-norm on a topological vector space V is continuous if and only if it is continuous at the origin.*

▷ **Exercise 11–2.** Prove the above proposition.

11.1.3 Proposition. *If $\| \; \|$ is a norm for the vector space V, then $\rho(v_1, v_2) := \|v_1 - v_2\|$ defines a metric for V, and with the topology defined by this metric V is a topological vector space.*

▷ **Exercise 11–3.** Prove this.

Remark. Such a topological vector space, with a fixed choice of norm, is called a *normed space*, and if the corresponding metric space is complete then it is called a *Banach space*. When there are several normed spaces in a given discussion it is customary to use the same symbol, $\| \; \|$, for the norm in all of them, and only use subscripts etc. to distinguish between the norms when there is danger of confusion.

11.1.4 Proposition. *If V is a normed space and p is a continuous semi-norm on V then there is a constant $K \geq 0$ such that $p(v) \leq K \|v\|$, and K is a Lipschitz constant for p.*

Proof. Since p is continuous at 0, there exists a $\delta > 0$ such that $\|v\| \leq \delta$ implies $p(v) \leq 1$ By Positive Homogeneity, if $v \neq 0$, then $\left\| \frac{\delta}{\|v\|} v \right\| = \delta$, so $\frac{\delta}{\|v\|} p(v) = p(\frac{\delta}{\|v\|} v) \leq 1$, and hence $p(v) \leq \frac{1}{\delta} \|v\|$, and we can take $K = \frac{1}{\delta}$. Then $|p(v_1) - p(v_2)| \leq p(v_1 - v_2) \leq K \|v_1 - v_2\|$.

Definition. If V and W are normed spaces and $T : V \to W$ is linear, then T is called *bounded* if there exists a $K \geq 0$ such that $\|Tv\| \leq K \|v\|$ for all $v \in V$. In this case we define the norm of T, denoted by $|||T|||$, to be the least such K; i.e., $|||T||| := \inf\{K \mid \|Tv\| \leq K \|v\| \text{ for all } v \in V\}$. We denote by $L(V, W)$ the set of all bounded linear maps of V into W.

$\triangleright$ **Exercise 11–4.** Show that $|||T||| := \sup_{v \neq 0} \frac{\|Tv\|}{\|v\|} = \sup_{\|v\|=1} \|Tv\|$.

$\triangleright$ **Exercise 11–5.** If T is as above then it is clear that if T is bounded then it is continuous and in fact $|||T|||$ is a Lipschitz constant for T. Conversely, show that if T is continuous at 0 then T is bounded. (Hint: one approach is to check that $v \mapsto \|Tv\|$ is a semi-norm for V and use the above proposition.)

$\triangleright$ **Exercise 11–6.** Show that if V and W are normed spaces then $L(V, W)$ is a vector space and $||| \; |||$ is a norm for $L(V, W)$. Show also that if W is a Banach space then so is $L(V, W)$.

Definition. Let V and W be normed spaces and $F \subseteq L(V, W)$. We say that F is *pointwise bounded* if for each $v \in V$, $\{Tv \mid T \in F\}$

is a bounded subset of W (i.e., $\sup\{||Tv|| \quad | \quad T \in F\} < \infty$), and that F is *uniformly bounded* if it is a bounded subset of $L(V, W)$ ((i.e., $\sup\{\ |||T|||\ |\ \ T \in F\} < \infty$).

The following result, with many important applications in analysis, demonstrates the power of the Baire Category Theorem.

11.1.5 Principle of Uniform Boundedness. *If V and W are Banach spaces then a pointwise bounded family F of continuous linear maps $T : V \to W$ is uniformly bounded.*

Proof. Since F is pointwise bounded, the union of the increasing sequence of closed sets, $E_n := \{v \in V \mid \|Tv\| \leq n \text{ for all } T \in F\}$ is all of V, so by the Baire Category Theorem some E_n has a non-empty interior, i.e., for some v_0 in V and $\epsilon > 0$, $\|v - v_0\| \leq \epsilon$ implies $\|Tv\| \leq n$. To prove the theorem it will suffice to find a bound of the form $\|Tv\| \leq M$ satisfied if $T \in F$ and $\|v\| \leq 1$, for this says that $|||T||| \leq M$ for all $T \in F$. Since F is pointwise bounded, $c = \sup_{T \in F} \|Tv_0\| < \infty$. And since $\|(v_0 + \epsilon v) - v_0\| \leq \epsilon$, $\|T(v_0 + \epsilon v)\| \leq n$ for all $T \in F$. Hence,

$$\epsilon \|Tv\| = \|T((v_0 + \epsilon v) - v_0)\| = \|T(v_0 + \epsilon v) - T(v_0)\| \leq n + c,$$

and we can take $M = \frac{n+c}{\epsilon}$. ∎

Definition. Two norms for the same vector space are called *equivalent* if they give the same topologies.

Remark. Suppose that $\|\ \|_1$ and $\|\ \|_2$ are two norms for the same vector space V, If we denote by V_1 and V_2 the two normed spaces we get by using these norms, and let $T : V_1 \to V_2$ denote the identity map, then $\|\ \|_1$ and $\|\ \|_2$ are equivalent if and only if T is bicontinuous. But since continuous implies bounded, this gives:

11.1.6 Proposition. *Two norms, $\|\ \|_1$ and $\|\ \|_2$, for the same vector space are equivalent if and only if there exist constants M_1 and M_2 such that for all $v \in V$, $\|v\|_1 \leq M_1 \|v\|_2$ and $\|v\|_2 \leq M_2 \|v\|_1$.*

11.1.7 Theorem. *If V is a finite dimensional vector space then any two norms for V are equivalent.*

Proof. This follows from the above remark since we have seen that linear maps between finite dimensional topological vector spaces are automatically continuous. ∎

Lecture 12. Compactness for Metric Spaces

A topological space X is called compact if it has the Heine-Borel Property; namely if every covering of X by open sets has a finite subcover. This turns out to be an extremely important property, and we will come back to study it more detail later in the setting of general topological spaces. In this section we will study compactness in the context of metric spaces, where it turns out to be possible to characterize in in several equivalent ways.

12.1. Some Properties Equivalent to Heine-Borel

▷ **Exercise 12–1.** A collection of subsets of a set X is said to have the *finite intersection property* (abbrev., f.i.p.) if every finite subcollection has non-empty intersection. Show that a space X is compact if and only if any collection of closed subsets of X with the finite intersection property has non-empty intersection. (Hint: De Morgan.)

Definition. Let S be a subset of a space X. A point $x \in X$ is called a *limit point* (or an *accumulation point*) of S if every neighborhood of x contains a point $s \in S$ with $s \neq x$, i.e., if x is adherent to $S \setminus \{x\}$

Definition. A metric space X is said to satisfy the *Bolzano-Weierstrass Property* if it has either and hence both of the following two equivalent properties:

1) Every sequence $\{x_n\}$ in X has a convergent subsequence.
2) Every infinite subset of X has a limit point.

Proof of Equivalence of 1) and 2).

Assume 1) is valid and let S be an infinite set. Choose a sequence $\{s_n\}$ of distinct points of S. Then by 1), by passing to a subsequence, we can assume that $\{s_n\}$ converges to a point $x \in X$. Since the s_n are distinct, $s_n \neq x$ for large n, and since $s_n \to x$ it follows that x is in the closure of $S \setminus \{x\}$, i.e., x is a limit point of S. So 1) implies 2).

Now assume 2) is valid and let $\{x_n\}$ be a sequence in X. One possibility is that there are only finitely many distinct x_n, in which case there is clearly a "constant" subsequence, i.e., one with all the elements are equal, and of course this is convergent. The other possibility is that there are infinitely many different x_n in which case, passing to a subsequence, we can assume that the x_n are distinct. Then by 2) there is a limit point x_0 of the set of x_n, so for every positive integer k we can find $x_{n_k} \neq x_0$ with $\rho(x_{n_k}, x_0) < \frac{1}{k}$, Since the x_n get arbitrarily close to x_0 there are infinitely many such x_k, so we can assume that $n_k > n_{k-1}$, and so $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$ which clearly converges to x_0.

Remark. Recall that a Cauchy sequence converges if it has a convergent subsequence, so if a metric space X satisfies the Bolzano-Weierstrass Property then any Cauchy sequence in X converges. That is:

12.1.1 Proposition. *A metric space X with the Bolzano-Weierstrass Property is complete.*

12.1.2 Proposition. *A metric space X that satisfies the Heine-Borel Property also satisfies the Bolzano-Weierstrass Property.*

Proof. (as improved by Zev Klagsbrun)

We must conclude from Heine-Borel that a subset S of X with no limit points is finite. If $x \in X$ then, since x is not a limit point of S, there is a neighborhood V_x of x such that $S \cap V_x$ is either empty (if $x \notin S$) or contains just the single point x. Since the V_x clearly cover X, by Heine-Borel there is a finite subcover of X, $\{V_{x_1}, \ldots, V_{x_n}\}$. Hence $S = S \cap \bigcup_{i=1}^{n} V_{x_i} = \bigcup_{i=1}^{n}(S \cap V_{x_i})$ is a subset of $\{x_1, \ldots, x_n\}$. ∎

12.1.3 Corollary. *A metric space X that satisfies the Heine-Borel Property is complete.*

Remark. Here is an alternate proof of the above Proposition. Let $\{x_n\}$ be a sequence in a metric space X that satisfies the Heine-Borel Property. If we define $F_n :=$ the closure of $\{x_k \mid k \geq n\}$, then $\{F_n\}$ is a decreasing sequence of non-empty closed sets so they have the finite intersection property, and so by the dual form of Heine-Borel there exists a point x_0 belonging to all the F_n. Note that this means that the distance of x_0 from $\{x_k \mid k \geq n\}$ is zero. It is now easy to choose inductively an increasing sequence n_k of positive integers such that $\rho(x_{n_k}, x_0) < \frac{1}{k}$, so $\{x_{n_k}\}$ is a convergent subsequence of $\{x_n\}$.

Lecture 13.
Compactness for Metric Spaces. (Cont.)

13.1. Lebesgue Numbers

Definition. If $\mathcal{C}$ is an open cover of a metric space X, then $\lambda > 0$ is called a *Lebesgue number* for $\mathcal{C}$ if every subset S of X with $\operatorname{diam}(S) < \lambda$, is included in some $O \in \mathcal{C}$.

13.1.1 Lebesgue Number Lemma. *If a metric space X has the Bolzano-Weierstrass property, then every open cover of X has a Lebesgue number.*

Proof. Suppose $\mathcal{C}$ is an open cover of X without any Lebesgue number. We will construct a sequence $\{x_n\}$ in X with no convergent subsequence, contradicting the Bolzano-Weierstrass property. For each positive integer n, since $\frac{1}{n}$ is not a Lebesgue number for $\mathcal{C}$, we can choose a set S_n of diameter less than $\frac{1}{n}$ such that S_n is not included in any $C \in \mathcal{C}$, and we choose $x_n \in S_n$. If a subsequence x_{n_k} converged to a point x_0, then since $\mathcal{C}$ is a cover, $x_0 \in C$ for some $C \in \mathcal{C}$, and since C is open, for some $\epsilon > 0$, $B_\epsilon(x_0) \subseteq C$. Now choose a positive integer k so large that $\rho(x_{n_k}, x_0) < \frac{\epsilon}{2}$ and also $\frac{1}{n_k} < \frac{\epsilon}{2}$. Then since $x_{n_k} \in S_{n_k}$ and $\operatorname{diam}(S_{n_k}) < \frac{1}{n_k} < \frac{\epsilon}{2}$ it follows that $S_{n_k} \subseteq B_\epsilon(x_0) \subseteq C$, contradicting that S_{n_k} was by choice not included in any element of $\mathcal{C}$. ∎

13.1.2 Corollary. *A continuous map f of a compact metric X into a metric space Y is uniformly continuous.*

Proof. If $\epsilon > 0$, the set of $B_{\frac{\epsilon}{2}}(y, Y)$ for all $y \in Y$ is an open cover of Y, so their inverse images $f^{-1}(B_{\frac{\epsilon}{2}}(y, Y))$ is an open cover of X. If δ is a Lesbesgue number for this latter cover, and $\rho_X(x_1, x_2) < \delta$, then $f(x_1)$ and $f(x_2)$ both belong to some $B_{\frac{\epsilon}{2}}(y, Y))$, hence $\rho_Y(f(x_1), f(x_2)) < \epsilon$.

Question. Can we strengthen the above corollary to say that the function f is not only uniformly continuous but even Lipschitz? No, and $f : [0.1] \to [0,1]$ defined by $f(x) := \sqrt{x}$ is a counter-example. Can you see why?

Recall that we saw we could always make a metric space bounded by replacing its metric ρ with the minimum of ρ and 1. Not only is this new metric equivalent to the original in the sense that it has the same open sets, but it even has the same Cauchy sequences. This makes it clear that bounded is not a topological property, but it turns out that a closely related property called totally bounded does have interesting and important topological consequences.

Definition. A metric space X is called *totally bounded* if for every $\epsilon > 0$, there exists a finite covering of X by sets of diameter less than ϵ. A subset of X is called totally bounded if it is totally bounded in the induced metric.

▷ **Exercise 13–1.** Show that a totally bounded metric space is bounded and give an example of a metric space that is bounded but not totally bounded.

Remark. In any metric space X, the collection of open $\frac{\epsilon}{2}$-balls centered at all the points of X is a covering of X by open sets of diameter less than ϵ. If X satisfies the Heine-Borel Property then this open cover will have a finite subcover. This proves that:

13.1.3 Proposition. *A metric space X that satisfies the Heine-Borel Property is totally bounded.*

Definition. If X is a metric space and $\epsilon > 0$, then a subset S of X is called *ϵ-dense in X* if every point of X is within ϵ of some point of S, i.e., if the collection of all open ϵ-balls centered at points of S covers X.

13.1.4 Proposition. *A metric space X is totally bounded if and only if for every $\epsilon > 0$, X has a finite ϵ-dense subset.*

▷ **Exercise 13–2.** Prove this. (Hint: an ϵ-ball has diameter $\leq 2\epsilon$.)

13.1.5 Proposition. *A metric space X is totally bounded if and only if every sequence in X has a Cauchy subsequence.*

Proof. Assume X is not totally bounded and choose $\epsilon > 0$ such that there is no finite ϵ-dense subset of X. Choose any $x_1 \in X$. Then since $B_\epsilon(x_1, X)$ does not cover X, we can choose $x_2 \notin B_\epsilon(x_1, X)$, and inductively we can choose $x_{n+1} \notin \bigcup_{1\leq j\leq n} B_\epsilon(x_j, X)$. Then since $\rho(x_n, x_m) \geq \epsilon$ for $n \neq m$, the sequence $\{x_n\}$ has no Cauchy subsequence.

For the converse, assume that X is totally bounded and let $\{x_n\}$ be any sequence in X. Cover X by a finite number of subsets of diameter less than 1. Then the sequence $\{x_n\}$ will be frequently in at least one of these sets, call it A_1. Next cover A_1 by a finite number of subsets of A_1 of diameter less than $\frac{1}{2}$, and choose one of these, A_2, such that $\{x_n\}$ is frequently in A_2. Inductively, cover A_k by a finite number of subsets of A_k having diameter less than $\frac{1}{k+1}$ and choose one of these, A_{k+1}, such that $\{x_n\}$ is frequently in A_{k+1}. In this way we get a nested sequence of sets A_k with $\operatorname{diam}(A_k) \to 0$ such that the sequence $\{x_n\}$ is frequently in A_k for every k. Now choose $x_{n_1} \in A_1$ and inductively choose $n_{k+1} > n_k$ such that $x_{n_k} \in A_k$. Since the $\operatorname{diam}(A_k) \to 0$, it is clear that $\{x_{n_k}\}$ is a Cauchy subsequence of $\{x_n\}$. ∎

13.1.6 Proposition. *A metric space X is complete and totally bounded if and only if it satisfies the Bolzano-Weierstrass Property.*

Proof. Immediate from the preceding Proposition and 12.1.1. ∎

Lecture 14.
Compactness for Metric Spaces, The End.

14.0.1 Theorem. *If X is a metric space then the following three properties of X are equivalent:*

1) X is compact, i.e., it has the Heine-Borel Property.

2) X is complete and totally bounded.

3) X has the Bolzano-Wierstrass Property.

Proof.

1) $\implies$ 2) is Proposition 12.1.3 and 13.1.2.

2) $\iff$ 3) is Proposition 13.1.6.

It remains to prove 2) $\implies$ 1). Assuming that X is a complete and totally bounded metric space and that $\mathcal{C}$ is an open cover X with no finite subcover, we will derive a contradiction. Since X is totally bounded, it can be covered by finitely many closed sets of diameter ≤ 1, and at least one of these, call it X_1 cannot be covered by any finite number of sets from $\mathcal{C}$. Define inductively a nested sequence of closed sets X_n such that the diam$(X_n) \leq \frac{1}{n}$ and no finite subset of $\mathcal{C}$ covers X_n. The inductive step works because, by total boundedness, we can cover X_n by a finite number of closed sets of diameter $< \frac{1}{n+1}$, and since no finite number of sets from $\mathcal{C}$ cover X_n, the same must be true of one of these closed sets, and we call it X_{n+1}. Since X_n is a nested sequence of non-empty closed sets in a complete metric and diam$(X_n) \to 0$, by Ascoli's Lemma, $\bigcap_n X_n$ is a single point $p \in X$. Since $\mathcal{C}$ is a covering of X, $p \in O$ for some $O \in \mathcal{C}$, and since O is open, $B_\epsilon(p, X) \subseteq O$ for some $\epsilon > 0$. Since X_k has diameter $< \frac{1}{k}$, it follows that $X_k \subseteq B_\epsilon(p, X) \subseteq O$ for $\frac{1}{k} < \epsilon$, contradicting that no finite number of sets of $\mathcal{C}$ cover any X_k. ∎

14.1. Uniform Continuity Extension Theorem

What is the point to singling out uniformly continuous functions? Do they have any nice properties not shared with just plain continuous functions? The following Lemma and Theorem provide a partial answer to that question.

14.1.1 Lemma. *If $f : X \to Y$ is uniformly continuous then it maps Cauchy sequences in X to Cauchy sequences in Y.*

Proof. Given $\epsilon > 0$, choose $\delta > 0$ so that $\rho_X(x_1, x_2) < \delta$ implies that $\rho_Y(f(x_1), f(x_2)) < \epsilon$. If $\{x_n\}$ is a Cauchy sequence in X then for m and n both sufficiently large, $\rho_X(x_n, x_m) < \delta$, hence $\rho_Y(f(x_n), f(x_m)) < \epsilon$, so $\{f(x_n)\}$ is a Cauchy sequence in Y. ∎

14.1.2 Uniform Continuity Extension Theorem. *Let D be a dense subspace of a metric space X and $f : D \to Y$ a uniformly continuous map of D into a complete metric space Y. Then f has a unique continuous extension to a map $\tilde{f} : X \to Y$ and $\tilde{f}$ is also uniformly continuous.*

Proof. Since D is dense, given x in X we can find a sequence of points $d_n \in D$ with $d_n \to x$. Since d_n is convergent it is Cauchy, so by the Lemma, $f(d_n)$ is a Cauchy sequence in Y and since Y is complete it converges, and we define $\tilde{f}(x) := \lim_{n\to\infty} f(d_n)$. In fact, for $\tilde{f}$ to be continuous we **must** define $\tilde{f}$ this way, so uniqueness is clear. But wait! How do we know that $\tilde{f}$ well-defined? That is, if d'_n is another sequence in D that also converges to x, why must $f(d'_n)$ have the same limit as $f(d_n)$? Well, the sequence $\{d''_n\}$ given by $\{d_1, d'_1, d_2, d'_2, \dots\}$ also converges to x so the sequence $f(d''_n)$ also has a limit, and since $f(d_n)$ and $f(d'_n)$ are subsequences of it they must have the same limit. Finally, given $\epsilon > 0$, choose δ such that if $d_1, d_2 \in D$ and $\rho_X(d, d') < \delta$ then $\rho_Y(f(d), f(d')) < \epsilon$. Given $x, x' \in X$ with $\rho_X(x, x') < \delta$, choose sequences d_n and d'_n in D converging to x and x' respectively. By the continuity of ρ_X, eventually $\rho_X(d_n, d'_n) < \delta$, so $\rho_Y(f(d_n), f(d'_n)) < \epsilon$, and by the continuity of ρ_Y, $\rho_Y(\tilde{f}(x), \tilde{f}(x')) < \epsilon$. ∎

Application: Quick and Dirty Riemann Integral.

As a simple application of the Extension Theorem that will suggest how powerful it is, we define a generalization of the Riemann integral for Banach space valued functions. Let V be a Banach space and $B([a, b], V)$ the Banach space of bounded functions from the interval $[a, b]$ into V. We note that $B([a, b], V)$ is a Banach space in the sup norm. Let $\mathcal{S} \subset B([a, b], V)$ denote the linear subspace of step functions, i.e., the space of $f \in B([a, b], V)$ that are constant on the subintervals $[x_i, x_{i+1})$ of some finite partition $a = x_0 < x_1 < \ldots < x_n = b$ of $[a, b]$, and we define a map $I : S \to V$ called the *integral* for such functions by $I(f) := \sum_{i=1}^{n}(x_i - x_{i-1})f(x_{i-1})$. Then it is clear that I is linear and that $\|I(f)\| \leq (b-a)\,\|f\|_\infty$, so I is Lipschitz, and hence uniformly continuous. By the above Extension Theorem. I extends uniquely to a continuous (and clearly linear) map $\tilde{I} : \bar{\mathcal{S}} \to V$. In particular, since the space $C([a, b], V)$ of all continuous maps $f : [a, b] \to V$ is clearly included in $\bar{S}$, this defines the integral on $C([a, b], V)$. For the special case $V = \mathbf{R}$, this is just the usual Riemann Integral.

What happens to the Extension Theorem if Y is not complete? Is there still some sort of unique extension of f to all of X? The answer is yes, but now the image will lie in "the completion of Y", a concept that we consider next

14.2. The Completion of a Metric Space

Definition. If X is a metric space, then a metric space Y is called a *completion of* X if X is a dense subspace of Y and Y is complete.

Example. The real field $\mathbf{R}$ is a completion of the field $\mathbf{Q}$ of rationals.

It is immediate from the Uniform Continuity Extension Theorem that if Y_1 and Y_2 are two completions of the same space X then the identity map of X extends uniquely to an isometry of Y_1 with Y_2, so in this fairly strong sense there is at most one completion of a metric space X, and for that reason it is customary to speak of **the** completion of X.

But does the completion always exist? Yes, and in fact there are a number of ways to construct it. Gamelin and Greene describe one standard construction (equivalence classes of Cauchy sequences) in Exercise 7, page 13 (and below we will use a variant of that method to construct the reals $\mathbf{R}$ from the rationals $\mathbf{Q}$), but the following is a somewhat simpler construction.

Notice that to show that X has a completion, it suffices to find a complete space W that includes X as a subspace, for then the closure of X in W will be a closed subspace of the complete space W and hence complete, and of course X is dense in its closure. This means it will suffice to find an isometry F of X into a complete space W, for then we can think of F as "embedding" X into W, i.e., we identify $x \in X$ with its image $F(x) \in W$. For W we will take the vector space $BC(X, \mathbf{R})$ of bounded, continuous real-valued functions on X, $f : X \to \mathbf{R}$. We know, this is a Banach space (hence complete) with respect to the sup norm: $\|f\|_\infty := \sup\{|f(x)| \mid x \in X\}$. We define the embedding $F : X \to BC(X, \mathbf{R})$ by choosing an arbitrary point $a \in X$ and defining $F(x)(x') := \rho(x', x) - \rho(x', a)$. It follows easily from the triangle inequality that $|F(x)(x')| \leq \rho(x, a)$ so $F(x)$ is bounded for each $x \in X$, and hence F maps X into $BC(X, \mathbf{R})$.

It remains to show that F is an isometry, i.e., that for all $x_1, x_2 \in X$, $\|F(x_1) - F(x_2)\|_\infty = \rho(x_1, x_2)$. By the triangle inequality

$$|F(x_1)(x) - F(x_2)(x)| = |\rho(x, x_1) - \rho(x, x_2)| \leq \rho(x_1, x_2),$$

holds for any $x \in X$, so $\|F(x_1) - F(x_2)\|_\infty \leq \rho(x_1, x_2)$. However,

$$|F(x_1)(a) - F(x_2)(a)| = |\rho(a, x_1) - \rho(a, x_2)| \geq \rho(x_1, x_2),$$

so we also have the reverse inequality $\|F(x_1) - F(x_2)\|_\infty \geq \rho(x_1, x_2)$, demonstrating that F is an isometry. We have proved:

14.2.1 Existence Theorem for Completions. *Every metric space has a completion, which is unique to within unique isometry.*

14.2.2 Corollary. *Let D be a dense subspace of a metric space X and $f : D \to Y$ a uniformly continuous map of D into a metric space Y. Then f has a unique continuous extension to a map $\tilde{f}$ of X into the completion of Y, and moreover $\tilde{f}$ is also uniformly continuous.*

Lecture 15.
The Field of Reals and its Compact Subsets

15.1. The Ordered Field of Real Numbers

You perhaps have seen a development of the real number system, $\mathbb{R}$, and a proof of its important properties (e.g., the Least Upper Bound Principle) starting from the rational numbers $\mathbb{Q}$ or even the integers. But it is interesting to revisit this vital piece of mathematics and look at it in our current context.

First some preliminary definitions. A ring $\mathcal{R}$ is said to be an ordered ring if there is a subset $\mathcal{P}$ of $\mathcal{R}$, called the set of *positive elements* that is closed under addition and multiplication, and satisfies the "trichotomy" condition; that for every element $x \in \mathcal{R}$ exactly one of the following holds, namely: $x = 0$, or $x \in \mathcal{P}$, or $-x \in \mathcal{P}$. (In these three mutually exclusive cases, we define $|x|$ to be 0, x or $-x$ respectively.) We then define a relation $<$ on $\mathcal{R}$ by $x < y$ if and only if $y - x \in \mathcal{P}$, and it follows easily that this is a linear ordering of $\mathcal{R}$ with the well-known properties. In particular, since $x^2 = (-x)^2$ it follows that if $x \neq 0$ then $x^2 \in \mathcal{P}$, i.e., $x^2 > 0$. In particular, if $\mathcal{R}$ has a unit, 1, then $1 = 1^2$ is positive, hence $1 + 1 + \ldots + 1 \neq 0$. It follows that an ordered field $\mathcal{F}$ always has characteristic zero, and so includes $\mathbb{Q}$ as its prime subfield. If $\{f_n\}$ is a sequence in $\mathcal{F}$ we say it is a Cauchy sequence if given any $\epsilon > 0$ in $\mathcal{F}$, eventually $|f_n - f_m| < \epsilon$ and we say f_n converges to $f \in \mathcal{F}$ if given any $\epsilon > 0$ eventually $|f_n - f| < \epsilon$. The usual arguments shows that limits are unique and that convergent implies Cauchy, and we say that $\mathcal{F}$ is a *complete ordered field* if every Cauchy sequence in $\mathcal{F}$ is convergent.

Definition. The ordered field $\mathcal{F}$ is called *archimedean* if the sequence $\frac{1}{n}$ converges to zero..

▷ **Exercise 15–1.** Prove that an ordered field $\mathcal{F}$ is archimedean if and only if the set $\mathbf{Z}^+$ of positive integers is not bounded above by any element of $\mathcal{F}$. (Hint: First show that if $0 < a < b$ then $0 < \frac{1}{b} < \frac{1}{a}$.)

Definition. The ordered field $\mathcal{F}$ is said to satisfy the *Least Upper Bound Principle* if every non-empty subset $S \subseteq \mathcal{F}$ that has an upper bound has a least upper bound (denoted as usual by $\sup S$).

15.1.1 Lemma. *If an ordered field $\mathcal{F}$ satisfies the Least Upper Bound Principle a monotonically increasing sequence in $\mathcal{F}$ that is bounded above converges to its least upper bound.*

▷ **Exercise 15–2.** Prove the Lemma.

Remark. If $\mathcal{F}$ satisfies the Least Upper Bound Principle and a non-empty set $S \subseteq \mathcal{F}$ is bounded below, then $-S$ is bounded above, so the greatest lower bound of S exists and is given by $\inf S = -\sup(-S)$. And of course a monotonically decreasing sequence in $\mathcal{F}$ that is bounded below converges to its greatest lower bound.

Remark. If $\mathcal{F}$ satisfies the Least Upper Bound Principle and a sequence f_n of elements of $\mathcal{F}$ is bounded below by A and above by B, then $\inf\{f_k \mid n \geq k\}$ is a monotonically increasing sequence that is also bounded above by B so it converges to a limit

$$\underline{\lim}\ f_n = \liminf_{n\to\infty} f_n := \lim_{n\to\infty} \inf\{f_k \mid n \geq k\},$$

and it is easy to see that $\underline{\lim}\ f_n$ is a limit point of the sequence f_n, and in fact that it is the smallest limit point. Similarly, $\sup\{f_k \mid n \geq k\}$ is a decreasing sequence that is bounded below by A and

$$\overline{\lim}\ f_n = \limsup_{n\to\infty} f_n := \lim_{n\to\infty} \sup\{f_k \mid n \geq k\}$$

is the largest limit point of f_n. Thus the sequence f_n converges if and only if $\underline{\lim}\ f_n = \overline{\lim}\ f_n$. In any case there exist subsequences that converge to $\underline{\lim}\ f_n$ and to $\overline{\lim}\ f_n$, so:

15.1.2 (Classical Bolzano-Weierstrass Theorem). *If an ordered field $\mathcal{F}$ satisfies the Least Upper Bound Principle then any bounded sequence in $\mathcal{F}$ has a convergent subsequence.*

Since a Cauchy sequence is clearly bounded (and converges if it has a convergent subsequence):

15.1.3 Corollary. *An ordered field $\mathcal{F}$ that satisfies the Least Upper Bound Principle is complete.*

15.1.4 Proposition. *An ordered field $\mathcal{F}$ that satisfies the Least Upper Bound Principle is archimedean.*

Proof. If not then the set $\mathbf{Z}^+$ of positive integers is bounded above and hence has a least upper bound M. Since $M - \frac{1}{2}$ is less than M it is not an upper bound for $\mathbf{Z}^+$, i.e., there is a positive integer n with $n > M - \frac{1}{2}$. But then $n + 1$ is a positive integer greater than $M + \frac{1}{2}$ and hence greater than M, a contradiction. ∎

15.1.5 Theorem. *An ordered field $\mathcal{F}$ satisfies the Least Upper Bound Principle if and only if it is complete and archimedean.*

Proof. One direction follows from the above corollary and proposition, so we must show that if $\mathcal{F}$ is complete and archimedean and S is a non-empty bounded subset of $\mathcal{F}$ then S has a least upper bound.

Choose $s_0 \in S$ and let B_0 be an upper bound for S. We construct an increasing sequence $s_n \in S$ and a decreasing sequence B_n of upper bounds for S inductively as follows. Let M_n be the midpoint $\frac{B_n - s_n}{2}$ of the interval $[s_n, B_n]$. Then if M_n is an upper bound for S we define $B_{n+1} := M_n$ and define $s_{n+1} := s_n$; otherwise we choose $s_{n+1} \in S$ with $s_n + 1 > M_n$ and define $B_{n+1} := B_n$. In either case, $B_{n+1} - s_{n+1} \leq \frac{B_n - s_n}{2}$ and hence $B_n - s_n \leq \frac{B_0 - s_0}{2^n}$. Since $\mathcal{F}$ is archimedean, $\frac{1}{2^n} \to 0$ in $\mathcal{F}$, hence the diameters of the nested sequence of closed intervals $[s_n.B_n]$ approaches zero, and by Ascoli's Lemma there is a unique point M in their intersection. Since the sequence B_n of upper bounds for S has M

as a limit, M is an upper bound for S, and since the sequence of points s_n of S converge to M, it is a least upper bound. ∎

Construction of R.

To construct **R** we start with the ordered field $\mathbb{Q}$ of rational numbers and "complete" it. But we must be careful—we cannot use the method of the preceding section, because that involved the space $BC(X, \mathbb{R})$, so proceeding that way would be circular. However, completing $\mathbb{Q}$ directly is easy and also rather pretty. We first construct the set $\mathcal{C}(\mathbb{Q})$ of all Cauchy sequences in $\mathbb{Q}$, and make it into a ring by defining addition and multiplication by $\{x_n\} + \{y_n\} = \{x_n + y_n\}$ and $\{x_n\}\{y_n\} = \{x_n y_n\}$. We note that $\mathbb{Q}$ is included as the subring of constant sequences and $1 \in \mathbb{Q}$ is a unit for $\mathcal{C}(\mathbb{Q})$. If a Cauchy sequence $\{q_n\}$ does not converge to 0, then $\{\frac{1}{q_n}\}$ is also Cauchy, and it follows that the subring $\mathcal{Z}$ of rational sequences that converge to zero is a maximal ideal of $\mathcal{C}(\mathbb{Q})$, so the quotient ring $\mathcal{C}(\mathbb{Q})/\mathcal{Z}$ is a field, This becomes an ordered field if we define an element to be positive if a representative Cauchy sequence of rationals q_n is eventually positive. The ordered field we get in this way is the field **R** of real numbers.

▷ **Exercise 15–3.** Prove that if we define **R** in this way then it is complete and archimedean.

▷ **Exercise 15–4.** Prove that a bounded sequence in $\mathbf{R}^n$ has a convergent subsequence. (Hint: First use the Classical Bolzano-Weierstrass Theorem to choose a subsequence making the first component convergent, then...). Deduce from this that bounded subsets of $\mathbf{R}^n$ are totally bounded. Hence:

15.1.6 Theorem. *A subset of $\mathbf{R}^n$ is compact if and only if it is closed and bounded.*

Remark. Is there an ordered field that is not archimedean? Consider the polynomial ring $\mathbb{Q}[X]$ and define $F(X)$ to be positive if and only if its leading coefficient is positive. Then the field of rational functions of

X can be ordered by defining $P(X)/Q(X)$ to be positive if P and Q are both positive or $-P$ and $-Q$ are both positive. In this ordered field X is an upper bound for $\mathbf{Z}^+$.

Final Remark. The classical fact that every positive real number has a square root means we can define $\mathcal{P}$ of positive elements intrinsically from its algebra as $\mathcal{P} := \{x^2 \mid x \in \mathbf{R}\}$. It follows that any automorphism A of $\mathbf{R}$ preserves $\mathcal{P}$ and hence is order preserving. Since A must be the identity on $\mathbb{Q}$, and any real number is the least upper bound of the rationals less than it, it follows that A is the identity. In other words, not only are any two complete archimedean ordered fields isomorphic, they are even uniquely isomorphic.

Lecture 16.
Product Metric Spaces and The Ascoli-Arzela Theorem

16.1. Products of Metric Spaces

We have not yet formally discussed how to make the product $\Pi_{i=1}^n X_i$ of metric spaces X_i into a metric space. The details are straightforward and easy. If $x \in \Pi_{i=1}^n X_i$, then $x = (x_1, \dots, x_n)$ where $x_i \in X_i$ is called the i-th component of x. The map $\Pi_i : \Pi_{i=1}^n X_i \to X_i$, defined by $\Pi_i(x) := x_i$ is called the i-th projection map. There are three common metrics to use for the product space, namely:

1) $\rho_{\max}(x^1, x^2) := \max_{1 \le i \le n} \rho_{X_i}(x_i^1, x_i^2)$
2) $\rho_1(x^1, x^2) := \sum_{1 \le i \le n} \rho_{X_i}(x_i^1, x_i^2)$
3) $\rho_2(x^1, x^2) := \sqrt{\sum_{1 \le i \le n} \rho_{X_i}(x_i^1, x_i^2)^2}$

It is easy to see that these are all Lipschitz equivalent. For example it is clear that $\rho_1 \le n\,\rho_{\max}$ and $\rho_{\max} \le \rho_1$.

▷ **Exercise 16–1.** Find Lipschitz bounds for the other two pairs of metrics.

It follows that we can use any of these metrics for most purposes, since they not only define the same topology but also the same Cauchy sequences and lead to the same notions of uniform continuity and Lipschitz maps. We will usually prefer $\rho_{\max}$. The following trivial proposition characterizes the topology of the product in terms of the topology of the factors.

16.1.1 Proposition. *A sequence $\{x^k\}$ in $\Pi_{i=1}^n X_i$ is Cauchy (resp., converges to a point x^0) if and only if each component sequence $\{x_i^k\}$ is Cauchy (resp., converges to x_i^0). Equivalently, the projection maps $\Pi_i : \Pi_{i=1}^n X_i \to X_i$ are all uniformly continuous.*

16.1.2 Proposition. *The continuous image of a compact space is compact.*

▷ **Exercise 16–2.** Prove this. (Hint: Let X be compact and $f : X \to Y$ continuous. If O_α is an open cover of $f(X)$ then $f^{-1}(O_\alpha)$ is an open cover of X, so there is a finite subcover $f^{-1}(O_{\alpha_i}),\ i = 1 \dots n)$ ...

16.1.3 Theorem. *A product metric space $\Pi_{i=1}^n X_i$ is compact if and only if each of the factor spaces X_i is compact.*

Proof. Since each X_i is the continuous image of the product under the projection Π_i, if the product is compact then so is each X_i. Conversely, if each X_i is compact then given a sequence $\{x^k\}$ in $\Pi_{i=1}^n X_i$, we can choose a subsequence for which $\{x_1^k\}$ converges. From that subsequence we can select a subsequence for which $\{x_2^k\}$ also converges, etc. In this way we end up with a subsequence $\{x^{k_j}\}$ for which all the component sequences $\{x_i^{k_j}\}$ converge, and hence the subsequence $\{x^{k_j}\}$ itself converges ∎

16.2. The Ascoli-Arzela Theorem

In what follows X is a compact metric space and Y a complete metric space, As usual $C(X, Y)$ denotes the space of continuous maps of X into Y. Since X is compact, every continuous $f : X \to Y$ is automatically bounded, so we get a metric ρ_∞ for $C(X, Y)$ defined by $\rho_\infty(f_1, f_2) := \sup\{\rho_Y(f_1(x), f_2(x)) \mid x \in X\}$. We have seen that $C(X, Y)$ is complete and that convergence in $C(X, Y)$ means uniform convergence. In many situations it is important to be able to choose a convergent sequence from a sequence of functions that belong to a subset $S \subseteq C(X, Y)$, which means we want to know under what conditions S is totally bounded. This is the question answered by the Ascoli-Arzela Theorem. Before stating and proving it we need to develop some notation, definitions, and lemmas.

Notation.

Ev : $C(X, Y) \times X \to Y$ denotes the evaluation map, $(f, x) \mapsto f(x)$. If $S \subseteq C(X, Y)$ and $x \in X$ then $S(x)$ denotes the image of $S \times \{x\}$ under Ev, i.e., $S(x) := \{f(x) \mid f \in S\}$.

16.2.1 Lemma. *The evaluation map $C(X, Y) \times X \to Y$ is continuous. Hence if S is a compact subset of $C(X, Y)$ then for each $x \in X$, $S(x)$ is a compact subset of Y.*

▷ **Exercise 16–3.** Prove this lemma. Hint: it says that if f_n converges uniformly to f and x_n converges to x, then $f_n(x_n)$ converges to $f(x)$.

Definition. If S is a subset of $C(X, Y)$ and $x_0 \in X$, we say that S *is equicontinuous at* x_0 if for every $\epsilon > 0$ there exists a $\delta > 0$ such that if $x \in X$ and $\rho_X(x, x_0) < \delta$, then $\rho_Y(f(x), f(x_0)) < \epsilon$ for all $f \in S$. We say that S *is equicontinuous* if it is equicontinuous at all points of X. Finally, we say that S *is uniformly equicontinuous* if for every $\epsilon > 0$ there exists a $\delta > 0$ such that if $x_1, x_2 \in X$ satisfy $\rho_X(x_1, x_2) < \delta$, then $\rho_Y(f(x_1), f(x_2)) < \epsilon$ for all $f \in S$.

▷ **Exercise 16–4.** Show that if $S \subseteq C(X, Y)$ and all f in S satisfy a Lipschitz condition $\rho_Y(f(x_1), f(x_2)) \leq K\rho_X(x_1, x_2)$ with a fixed Lipschitz constant K then S is uniformly equicontinuous.

▷ **Exercise 16–5.** Show that if $S \subseteq C([a, b], \mathbf{R})$ and if all f in S are continuously differentiable and for some fixed $K > 0$ satisfy $|f'(x)| \leq K$ for $a \leq x \leq b$ then S is uniformly equicontinuous.

16.2.2 Ascoli-Arzela Theorem. *If Y is any metric space and X is a compact metric space, then a subset S of $C(X, Y)$ is totally bounded if and only if:*

1) *S is uniformly equicontinuous, and*
2) *For each $x \in X$, $S(x)$ is a totally bounded subset of Y.*

Before starting the proof we note that we can assume that Y is complete, since otherwise we can just replace Y by its completion. Similarly we can assume that S is closed in $C(X, Y)$, for otherwise we can replace S by its closure, since $\bar{S}$ is totally bounded if and only if S is. Then since $C(X, Y)$ is complete, so is S, and what we must show is that S is compact if and only if it is equicontinuous and $S(x)$ is compact for each x in X.

Proof. If S is compact then by the above Lemma each $S(x)$ is compact, and given $\epsilon > 0$ we can find an $\frac{\epsilon}{3}$-dense subset $f_1, \dots, f_n$ of S. Choose δ_i such that $\rho(x_1, x_2) < \delta_i$ implies $\rho(f_i(x_1), f_i(x_2)) < \frac{\epsilon}{3}$ and let δ be the minimum of these δ_i. If $f \in S$, choose f_i within $\frac{\epsilon}{3}$ of f. Then

$$\begin{aligned}\rho(f(x_1), f(x_2)) &\leq \rho(f(x_1), f_i(x_1)) + \rho(f_i(x_1), f_i(x_2)) + \rho(f_i(x_2), f(x_2)) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} < \epsilon,\end{aligned}$$

proving that S is uniformly equicontinuous.

Now suppose conversely that S is uniformly equicontinuous and each $S(x)$ is compact. Given $\epsilon > 0$, choose $\delta > 0$ such that $\rho(x_1, x_2) < \delta$ implies $\rho(f(x_1), f(x_2)) < \frac{\epsilon}{3}$ for all $x_1, x_2, \in X$ and $f \in S$, and choose $x_1, x_2, \dots, x_k$ to be δ-dense in X. Then $S(x_1) \times \cdots \times S(x_k)$ is a compact subset in Y^k, so its subset $\mathcal{F} = \{(f(x_1), f(x_2), \dots, f(x_k)) \mid f \in S\}$ is a totally bounded subset of Y^k. Select $f_1, f_2, \dots f_m \in S$ so that $(f_i(x_1), f_i(x_2), \dots, f_i(x_k)),\ i = 1 \dots, m$ is $\frac{\epsilon}{3}$-dense in $\mathcal{F}$. We will complete the proof by showing that $f_1, f_2, \dots f_m$ is ϵ-dense in S. We have to show that given $f \in S$ there is an f_j satisfying $\rho(f(x), f_j(x)) < \epsilon$ for all $x \in X$. Choose f_j with $\rho(f(x_i), f_j(x_i)) < \frac{\epsilon}{3}$ for $i = 1, \dots, k$. If $x \in X$ choose x_i with $\rho(x, x_i) < \delta$. Then $\rho(f(x), f_j(x)) \leq \rho(f(x), f(x_i)) + \rho(f(x_i), f_j(x_i)) + \rho(f(x), f_j(x)) \leq \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3}$. ∎

▷ **Exercise 16–6.**

a) Assume that X and Y are compact metric spaces. Show that the set of all maps $f : X \to Y$ that satisfy a Lipschitz condition with a fixed Lipschitz constant K is a compact subset of $C(X, Y)$.

b) If X is a compact metric space, show that the set $\mathrm{Isom}(X)$ of isometries of X with itself is a compact group.

c) Show that the set of continuously differentiable maps $f : [a, b] \to \mathbf{R}$ such that $|f(x)| \leq M$ and $|f'(x)| \leq K$ for fixed constants M and K and all $x \in [a, b]$ is a compact subset of $C([a, b], \mathbf{R})$.

16.3. Dini's Theorem

Recall that if X is a space and Y a metric space, then the uniform limit of a sequence of continuous functions from X into Y is also continuous. Dini's Theorem is a useful but very partial converse of this result. In case X is compact and $Y = \mathbf{R}$, it says that if a sequence of continuous functions converges pointwise to a continuous function then the convergence is automatically uniform *provided it is monotonic*.

16.3.1 Dini's Theorem. *Let X be a compact space an let f_n be a sequence of continuous real-valued functions on X. Assume that for each $x \in X$ the sequence $f_n(x)$ is monotonically decreasing and bounded below, and let $f(x)$ be its limit. Then if f is continuous, f_n converges uniformly to f.*

Proof. Since f is continuous, by replacing f_n by $f_n - f$ we can assume that f is identically zero. Given $\epsilon > 0$, let O_n denote the open set where f_n is less than ϵ. To prove that f_n converges uniformly to zero we must show that $O_n = X$ for n sufficiently large. But since f_n is monotonically decreasing, O_n is an increasing sequence of open sets, and since $f_n(x)$ converges pointwise to 0 the union of the O_n is all of X, hence by Heine-Borel some O_n must be all of X. ∎

Remark. Of course there is an analogous theorem in case the f_n are monotonically increasing rather than decreasing.

Lecture 17.
Topological Spaces

17.1. Topologies for a Set.

In the first lecture of the course we gave, informally, the definition of a topological space (or simply a “space”) and some related basic concepts. Here they are again more formally.

Definitions. If X is a set then a collection $\mathcal{T}$ of subsets of X is called a *topology* for X if

1) it is closed under arbitrary unions and finite intersections, and
2) it contains X and the empty set $\emptyset$.

A *topological space* consists of a set X together with a fixed topology $\mathcal{T}$ for X. The elements of $\mathcal{T}$ are called the open sets of X, and their complements are called the closed sets of X. If S is a subset of X then the interior of S is the largest open set included in S, i.e., the union of all open sets included in S, and the closure of S is the smallest closed set that includes S, i.e., the intersection of all closed sets that include S. If $x \in X$, then a subset N of X is called a neighborhood of x in X if x is contained in the interior of N.

$\triangleright$ **Exercise 17–1.** If X is a space then a collection $\mathcal{B}$ of open sets is called a *base* (or *basis*) for the topology of X if every open set is the union of sets from $\mathcal{B}$. If X is a set, show that a collection $\mathcal{B}$ of subsets of X is a base for some topology $\mathcal{T}$ for X if and only if:

1) Every point of X is contained in some $B \in \mathcal{B}$.
2) if $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$ then there exists a $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$.

The most common type of topological space is of course a metric space. But the definition allows for much more. For example any set X

can be made into a topological space by letting $\mathcal{T}$ be the so-called "trivial topology" consisting of just X and $\emptyset$. Another "strange" topology is the so-called *cofinite topology* for a set X, which consists of $\emptyset$ together with the complements of finite sets.

In order to develop interesting and useful general theorems concerning topological spaces, it is necessary to put restrictions on the topology $\mathcal{T}$. We already have seen that a metric space topology satisfies an important extra axiom; namely the so-called Hausdorff condition, that distinct points have disjoint neighborhoods. This is just one of a list of so-called "separation axioms" that we consider next. To simplify the statements of these axioms we will say that two subsets S_1 and S_2 of a topological space X *can be separated by open sets* to mean that there are disjoint open sets O_1 and O_2 with $S_1 \subseteq O_1$ and $S_2 \subseteq O_2$.

Separation Axioms.

T1: A topological space is called a T1-space if given distinct points p and q there exists an open set that contains q but not p. Equivalently, a topological space is a T1-space if every point is a closed set.

T2: A topological space is called a T2-space or a *Hausdorff* space if any two distinct points can be separated by open sets.

T3: A topological space is called a T3-space or a *regular* space if it is a T1-space and if any closed set F and any point $x \notin F$ can be separated by open sets.

T3: A topological space is called a T4-space or a *normal* space if it is a T1-space and if any two disjoint closed sets can be separated by open sets.

▷ **Exercise 17–2.** Show that the two versions of the T1 axiom are equivalent.

▷ **Exercise 17–3.** Show that T4 $\implies$ T3 $\implies$ T2 $\implies$ T1.

Definition. If S is a subset of a topological space X, then the *relative topology* for S is defined by declaring its open subsets to be sets of the form $S \cap O$ where O is open in X.

▷ **Exercise 17–4.** Show that the relative topology satisfies the axioms for a topology, and that if X is compact then a closed subset S of X is compact in the relative topology. Hint: If $U_\alpha = S \cap O_\alpha$ is an open cover of S then the O_α together with $\complement S$ is an open cover of X.

▷ **Exercise 17–5.** Show that if X is a Hausdorff space, then a compact subset K and any $p \notin K$ can be separated by open sets. Use this to prove the following proposition.

17.1.1 Proposition. *A compact subspace of a Hausdorff space X is closed in X.*

17.1.2 Proposition. *A compact Hausdorff space is normal.*

▷ **Exercise 17–6.** If X is a Hausdorff space, show that any two disjoint compact sets can be separated by open sets, and use this to prove the preceding proposition. (Hint: The special case where one of the two compact sets is a point has already already been handled in the preceding exercise.)

17.1.3 Proposition. *A metric space is normal.*

Proof. Let F_1 and F_2 be disjoint closed subsets in a metric spaces X. For each $x \in F_1$, $\delta_2^x := \frac{1}{3}\operatorname{dist}(x, F_2) > 0$ because F_2 is closed and $x \notin F_2$. Define $O_1 := \bigcup_{x \in F_1} B_{\delta_2^x}(x, X)$. Similarly define $O_2 = \bigcup_{x \in F_2} B_{\delta_1^x}(x, X)$ where $\delta_1^x := \frac{1}{3}\operatorname{dist}(x, F_1)$. Then it is easy to verify that F_1 and F_2 are separated by the open sets O_1 and O_2. ∎

Definition. If X and Y are topological spaces, a map $f : X \to Y$ is called a homeomorphism if f is a bijection and both f and f^{-1} are continuous.

17.1.4 Lemma. *If $f : X \to Y$ is a continuous bijection, then f is a homeomorphism if and only if it is a "closed mapping", i.e., F closed in X implies $f(F)$ is closed in Y.*

17.1.5 Proposition. *If X is a compact space and Y is Hausdorff, then a continuous bijection, $f : X \to Y$ is a homeomorphism.*

Proof. If F is closed in X, then since X is compact F is also compact, so since f is continuous, $f(F)$ is compact and since Y is Hausdorff, $f(F)$ is closed in Y. ∎

Lecture 18.
The Urysohn and Tietze Extension Theorems

18.1. The Theorems of Urysohn and Tietze

If S_0 and S_1 are disjoint subsets of a topological space X, there is an (apparently) stronger property than their being separated by open sets, namely that there exists a continuous function $f : X \to \mathbf{R}$ that is identically 0 on S_0 and identically one on S_1. In this case we say that S_0 and S_1 *can be separated by a continuous real-valued function*.

▷ **Exercise 18–1.** Why is this stronger than their being separated by open sets?

Note that we can also regard this property as a "continuous extension theorem". For the function defined on the subspace $S_0 \cup S_1$ that is zero on S_0 and one on S_1 is clearly continuous, and S_0 and S_1 can be separated by a continuous real-valued function if and only if this function can be extended to a continuous function on all of X.

There are two classic continuous extension theorems that are important enough that they should be part of the arsenal of every serious mathematician. One of these is:

18.1.1 Urysohn's Lemma. *Two disjoint closed subsets of a normal topological space can be separated by a continuous real-valued function.*

and the other is:

18.1.2 The Tietze Extension Theorem. *A continuous function $f : F \to [a, b]$ defined on a closed subspace F of a normal space X can be extended to a continuous function $g : X \to [a, b]$.*

We will consider these two non-trivial results below, where the strategies will be explained and liberal hints given, but you will be asked to provide the detailed arguments.

18.2. Urysohn's Lemma.

The definition of a normal space only mentions open and closed sets and says nothing about continuous functions. How can we connect these concepts? Here is the clue.

Let us define a *dyadic nest* in a space X to be a collection of open subsets $\{O_\alpha\}$ indexed by the dyadic rationals $\alpha = \frac{k}{2^n}$ in the open interval $(0, 1)$ such that whenever $0 < \alpha < \beta < 1$ are dyadic rationals, then $\overline{O}_\alpha$ (the closure of O_α) is included in O_β.

▷ **Exercise 18–2.** Show that if $f : X \to [0, 1]$ is continuous and for each dyadic rational $\alpha \in (0, 1)$ we define $O_\alpha := \{x \in X \mid f(x) < \alpha\}$ then $\{O_\alpha\}$ is a dyadic nest, $f^{-1}(0)$ is included in each O_α, and $f^{-1}(1)$ is disjoint from each O_α.

▷ **Exercise 18–3.** Show that if $\{O_\alpha\}$ is a dyadic nest in a space X and if we define $f : X \to [0, 1]$ by $f(x) := 1$ if x does not belong to any O_α and otherwise $f(x) := \inf\{\alpha \mid x \in O_\alpha\}$ then f is continuous. (Hint: You will need to use that the dyadic rationals are dense in $\mathbf{R}$, and also that continuity follows from $f^{-1}(-\infty, r)$ and $f^{-1}(r, \infty)$ being open for every $r \in \mathbf{R}$.)

The following Lemma is the key tool used in constructing a dyadic nest.

18.2.1 Lemma. *If X is a normal space and $F \subseteq U \subseteq X$ with F closed and U open, then there is an open set O with $F \subseteq O \subseteq \overline{O} \subseteq U$.*

Proof. Since F and $\complement U$ are disjoint closed sets, by normality we can choose disjoint open sets O and O' with $F \subseteq O$ and $\complement U \subseteq O'$. Since $O \subseteq \complement O' \subseteq U$ and $\complement O'$ is closed, $\overline{O} \subseteq U$. ∎

Lecture 19.
Urysohn and Tietze Theorems (Continued)

19.1. Proof of Urysohn's Lemma

We recall the lemma stated at the end of the last lecture and which we said was a key tool for construction dyadic nets

18.2.1 Lemma. If X is a normal space and $F \subseteq U \subseteq X$ with F closed and U open, then there is an open set O with $F \subseteq O \subseteq \overline{O} \subseteq U$.

Proof. Since F and $\complement U$ are disjoint closed sets, by normality we can choose disjoint open sets O and O' with $F \subseteq O$ and $\complement U \subseteq O'$. Since $O \subseteq \complement O' \subseteq U$ and $\complement O'$ is closed, $\overline{O} \subseteq U$. ∎

Now suppose we are given a normal space X and disjoint closed subsets S_0 and S_1 of X. By Exercise 18-3, to show that S_0 and S_1 can be separated by a continuous real-valued function (proving Urysohn's Lemma) it will suffice to construct a dyadic nest $\{O_\alpha\}$ such that for all $\alpha \in (0,1)$, $S_0 \subseteq O_\alpha \subseteq \complement S_1$. Here is the strategy.

Define $O_1 := \complement S_1$, so $S_0 \subseteq O_1$ and we can use the lemma with $F = S_0$ and $U = O_1$ to find an open O_0 with $S_0 \subseteq O_0 \subseteq \overline{O_0} \subseteq O_1 \subseteq \complement S_1$. Use the lemma a second time with $F = \overline{O_0}$ and $U = O_1$ to get $O_{\frac{1}{2}}$ with $\overline{O_0} \subseteq O_{\frac{1}{2}} \subseteq \overline{O_{\frac{1}{2}}} \subseteq O_1 \subseteq \complement S_1$. Next use the lemma two more times to get $O_{\frac{1}{4}}$ with $O_0 \subseteq O_{\frac{1}{4}} \subseteq \overline{O_{\frac{1}{4}}} \subseteq O_{\frac{1}{2}}$ and $O_{\frac{3}{4}}$ with $\overline{O_{\frac{1}{2}}} \subseteq O_{\frac{3}{4}} \subseteq \overline{O_{\frac{3}{4}}} \subseteq O_1$. Inductively, assuming we have used the lemma in this way to define the O_α for all $\alpha = \frac{k}{2^{n-1}}$ with $k = 0, 1, \dots 2^{n-1}$, we can then use the lemma n more times to get an $O_{\frac{k}{2^n}}$ with $\overline{O_{\frac{k-1}{2^n}}} \subseteq O_{\frac{k}{2^n}} \subseteq \overline{O_{\frac{k}{2^n}}} \subseteq O_{\frac{k+1}{2^n}}$ for k an odd integer between 0 and 2^n.

▷ **Exercise 19–1.** Complete the proof of Urysohn's Lemma by carrying out the details of this strategy.

19.2. Proof of the Tietze Extension Theorem.

▷ **Exercise 19–2.** Show that in the statement of Tietze's Theorem we can assume without loss of generality that $f : F \to [0, 1]$. (We will assume this in what follows.)

▷ **Exercise 19–3.** The Tietze theorem says that a continuous and *bounded* real-valued function defined on a closed subspace of a normal space X can be extended to a continuoud function on all of X, Show that the boundedness assumption is unnecessary. Hint: Think about the function $\tan(x)$.

We are assuming that X is normal, F is closed in X, and $f : F \to [0, 1]$ is continuous, and we want to construct a continuous $g : X \to [0, 1]$ that agrees with f on F. The strategy will be to do this in stages, using Urysohn's Lemma to construct inductively an infinite series of continuous functions with terms $g_n : X \to [0, 1]$ whose partial sums $\sum_{k=1}^{n} g_k$ converge uniformly to the desired g on all of X and to f on F.

We define inductively two sequences of functions, $f_n : F \to [0, (\frac{2}{3})^n]$ and $g_n : X \to [0, \frac{1}{3}(\frac{2}{3})^{n-1}]$

To start the induction we take $f_0 := f$ and use Urysohn's Lemma to construct $g_1 : X \to [0, \frac{1}{3}]$ such that $g_1(x) = 0$ if $x \in F$ and $f(x) \leq \frac{1}{3}$, and $g_1(x) = \frac{1}{3}$ if $x \in F$ and $f(x) \geq \frac{2}{3}$.

▷ **Exercise 19–4.** Explain how the existence of g_1 follows from Urysohn.

For the inductive step, assume that $f_0, \dots, f_{n-1}$ and $g_1, \dots, g_n$ have been defined with the domains and codomains given above. First define $f_n : F \to [0, (\frac{2}{3})^n]$ by $f_n(x) := f_{n-1}(x) - g_n(x)$.

▷ **Exercise 19–5.** Show that $0 \leq f_n(x) \leq (\frac{2}{3})^n$ for all $x \in F$.

To complete the inductive definitions, we again appeal to Urysohn to get $g_{n+1} : X \to [0, \frac{1}{3}(\frac{2}{3})^n]$ with $g_{n+1}(x) = 0$ if $x \in F$ and $f_n(x) \leq \frac{1}{3}(\frac{2}{3})^n$ and $g_{n+1}(x) = \frac{1}{3}(\frac{2}{3})^n$ if $x \in F$ and $f_n(x) \geq \frac{2}{3}(\frac{2}{3})^n$.

▷ **Exercise 19–6.** Show that the infinite series $\sum_{n=1}^{\infty} g_n$ converges uniformly to a function g that is therefore continuous on X. (Hint: M-Test.)

▷ **Exercise 19–7.** Show that this function g agrees with f on F. (Hint: Add the equalities $f - g_1 = f_1, f_1 - g_2 = f_2, \ldots, f_{n-1} - g_n = f_n$ to get $f - \sum_{k=1}^{n} g_k = f_n$ and note that f_n tends uniformly to zero.)

Lecture 20. Moore-Smith Convergence

Arguments using convergent sequences are a powerful and intuitive tool for investigating the topology of metric spaces, or more generally of first countable spaces. Showing that $x_n \to x$ implies $f(x_n) \to f(x)$ is often the easiest proof of continuity and showing that any sequence has a convergent subsequence is often the quickest route to showing compactness. A third example explains the efficacy of convergent sequences; namely a subset of a space is closed if and only if whenever a sequence of its points converges, the limit is in the subset. This shows that the convergent sequences determine the closed sets and hence the topology. However, outside the realm of first countable spaces this criterion for a subset being closed fails, and as a result sequences lose their power. This makes it natural to seek out some notion of "generalized sequences" to replace them, and in fact there are two well-known concepts that can replace sequences for non first countable spaces. One is the machinery of filters invented by Henri Cartan and developed by the Bourbaki group of French mathematicians. The other, and the one that we will study below, goes by the name of Moore-Smith Convergence or "directed sets and nets".

20.1. Directed Sets and Nets

Definition. A *directed set* is a partially ordered set D such that any two elements of D have an upper bound in D; i.e., if $d_1, d_2 \in D$ then there is a $d \in D$ with $d_1 \leq d$ and $d_2 \leq d$.

Of course any linearly ordered set is a directed set, so in particular the positive integers are. A very important example for us is the directed set $\mathcal{N}$ of neighborhoods of a point in a topological space, where if N and N' are in $\mathcal{N}$ we define $N \leq N'$ to mean that $N' \subseteq N$. **Note carefully that greater than means included in, not the other**

way around. So if $d_1, d_2 \in \mathcal{N}$ then we can take $d := d_1 \cap d_2$ as their upper bound.

▷ **Exercise 20–1.** Show that if A and B are directed sets then $A \times B$ becomes a directed set if we define $(a_1, b_1) \leq (a_2, b_2)$ to mean that $a_1 \leq a_2$ and $b_1 \leq b_2$.

Definition. If A is a directed set and X is any set, then a *net in X based on A* is a function $\alpha \mapsto x_\alpha$ from A into X. We will write $\{x_\alpha\}_{\alpha \in A}$ to denote net based on A. If S is a subset of X we say that x_α is *frequently in S* if for each $\alpha_0 \in A$ there is an $\alpha \geq \alpha_0$ with $x_\alpha \in S$, and we say that x_α is *eventually in S* if there is an $\alpha_0 \in A$ such that $x_\alpha \in S$ for all α with $\alpha \geq \alpha_0$.

Remark. It follows from the definition of a directed set that if x_α is eventually in S_1 and eventually in S_2, then it is also eventually in $S_1 \cap S_2$ (and in fact this is the motivation for the condition that each pair of elements in a directed set have an upper bound).

Definition. If X is a topological space and $\{x_\alpha\}$ is a net in X, then we call $x_0 \in X$ a *cluster point* of $\{x_\alpha\}$ if for each neighborhood N of x_0, x_α is frequently in N. We say that x_α *converges to* x_0 and write $x_\alpha \to x_0$ if for any neighborhood N of x_0, x_α is eventually in N. In this case we also say that x_0 is a *limit* of x_α. If x_0 is the unique limit of x_α then we write $\lim_{\alpha \in A} x_\alpha = x_0$.

We now generalize to net convergence some of the propositions proved earlier for sequential convergence in the metric case.

20.1.1 Proposition. *A topological space X is Hausdorff if and only if any two limits of a net in X are equal.*

▷ **Exercise 20–2.** Prove this proposition. Hint: If x and y cannot be separated by open sets, let $D := \{(U, V) \mid U \text{ and } V \text{ open}, x \in U, y \in V\}$. Define an ordering to make D a directed set and define a net based on D that converges to both x and y.

20.1.2 Proposition. *If X and Y are topological spaces then a function $f : X \to Y$ is continuous if and only if for every net $\{x_\alpha\}_{\alpha\in A}$ in X, $x_\alpha \to x_0$ implies $f(x_\alpha) \to f(x_0)$*

▹ **Exercise 20–3.** Prove this proposition.

Here is the proposition that says that knowledge of the limits of convergent nets determines the topology of a space.

20.1.3 Proposition. *If X is a topological space and S is a subset of X, then a point $x \in X$ is adherent to S if and only if there is a net in S that converges to x. Hence S is a closed subset of X if and only if whenever a net in S converges to a point $x \in X$ it follows that $x \in S$.*

▹ **Exercise 20–4.** Prove this proposition.

20.2. Subnets

Definition. If A and B are directed sets, a map $\alpha : B \to A$ is called *cofinal* if for every $\alpha_0 \in A$ there is a $\beta_0 \in B$ such that $\beta \geq \beta_0 \implies \alpha(\beta) \geq \alpha_0$.

Definition. If $\{x_\alpha\}_{\alpha\in A}$ is a net based on A and $\alpha : B \to A$ is cofinal, then the net based on B, $\{x_{\alpha(\beta)}\}_{\beta\in B}$ is called a *subnet* of $\{x_\alpha\}_{\alpha\in A}$.

20.2.1 Proposition. *A net $\{x_\alpha\}_{\alpha\in A}$ in a space X has a subnet that converges to a point $x \in X$ if and only if x is a cluster point of x_α, i.e., if and only if x_α is frequently in each neighborhood of x.*

▹ **Exercise 20–5.** Prove this proposition. Hint: If $\{x_\alpha\}_{\alpha\in A}$ is frequently in each neighborhood of x, let $\mathcal{N}$ be the set of neighborhoods of x made into a directed set as above. Let D be the subset of the directed set $A \times \mathcal{N}$ consisting of $\delta = (a, N)$ such that $x_a \in N$, and let $\alpha : D \to A$ be defined by $\alpha(a, N) := a$. Show that D is a directed set in the ordering induced from $A \times \mathcal{N}$, that $\alpha : D \to A$ is cofinal, and that the subnet $\{x_{\alpha(\delta)}\}_{\delta \in D}$ converge to x.

20.2.2 Proposition. *A space X is compact if and only if every net in X has a convergent subnet.*

$\triangleright$ **Exercise 20–6.** Prove this proposition. Hint: Assuming X is compact and $\{x_\alpha\}_{\alpha \in A}$ is a net in X, for each $\alpha \in A$ let $F_\alpha :=$ the closure of $\{x_\beta \mid \beta \geq \alpha\}$. Show that these F_α are closed sets with the finite intersection property, so that they have non-empty intersection. Then verify that any point in the intersection is a cluster point of x_α and use the preceding proposition. For the converse, assuming nets in X have cluster points and that $\mathcal{F}$ is a collection of closed sets with the finite intersection property, let D denote the directed set of finite intersections of sets from $\mathcal{F}$ ordered by (reverse) inclusion, and for each $\delta \in D$ choose $x_\delta \in \delta$, and show that any cluster point of $\{x_\delta\}_{\delta \in D}$ is in $\bigcap\{F \mid F \in \mathcal{F}\}$.

20.3. Universal Nets

So far everything has been straightforward generalization. Now we get to something interesting—and surprising.

Definition. A net $\{x_\alpha\}_{\alpha \in A}$ in a set X is called a *universal net for* X if for each subset S of X, either x_α is eventually in S or else x_α is eventually in $X \setminus S$, the complement of S.

The following obvious fact is part of the reason universal nets are useful.

20.3.1 Proposition. *A universal net converges if and only if it has a convergent subnet, or equivalently if and only if it has a cluster point.*

$\triangleright$ **Exercise 20–7.** Show that a sequence in X that is a universal net must be eventually constant. Hint: Assume that $\{x_n\}$ is a NECUS, i.e., a sequence that is a universal net and is not eventually constant. We want to derive a contradiction. First derive two properties, NECUS-1 and NECUS-2:

NECUS-1: If $S \subseteq X$ is finite then $x_n \in S$ for only finitely many n.

(Hint: If $s \in S$, then since $\{x_n\}$ is not eventually constant, x_n is not eventually in $\{s\}$ so, by universality, x_n is eventually in $X \setminus \{s\}$.)

NECUS-2: If $V \subseteq X$ is infinite and consists of values of x_n, then x_n is eventually in V.

(Hint: Otherwise x_n must be eventually in $X \setminus V$, and that would imply that V has only finitely many values of x_n.)

So, if x_n is a NECUS, it follows from NECUS-1 that the set V of values assumed by x_n is infinite. Partition V into two disjoint infinite subsets and use NECUS-2 to derive a contradiction.

(In fact, the only universal nets that you are likely to "see" without calling on the Axiom of Choice for help are the eventually-constant ones.)

20.3.2 Proposition. *If $\{x_\alpha\}_{\alpha \in A}$ is a universal net for X and if $f : X \to Y$ is any function, then $\{f(x_\alpha)\}_{\alpha \in A}$ is a universal net in Y.*

▷ **Exercise 20–8.** Prove this proposition.

Our next goal is to prove the important fact that every net has a universal subnet. This is a purely set theoretic result—no topology is involved—and in fact it is known to be equivalent to the Axiom of Choice. Nevertheless it is a key result in our approach to Tychonoff's Theorem—which is usually considered as one of the more important and difficult theorems of point set topology. The reason our proof of Tychonoff's Theorem will appear to be easy is that we have pushed the difficulties onto the proof of existence of universal subnets.

Remark. In studying the arguments below it is important to keep in mind the following elementary fact. If $\{x_\alpha\}$ is any net in X and $A \subseteq X$, then there are three exhaustive and mutually exclusive possibilities:

1) x_α is eventually in A, in which case it is **not** frequently in $X \setminus A$.
2) x_α is eventually in $X \setminus A$, in which case it is **not** frequently in A.

3) x_α is frequently in A **and** frequently in $X \setminus A$.

20.3.3 Lemma 1. *Let $\mathcal{C}$ be a collection of subsets of a set X that is directed by inclusion. If a net $\{x_\alpha\}_{\alpha\in A}$ in X is frequently in each member of $\mathcal{C}$, then some subnet is eventually in each member of $\mathcal{C}$.*

Proof. Order $D := \{(a, C) \in A \times \mathcal{C} \mid x_a \in C\}$, by $(a, C) \leq (a', C')$ if $a \leq a'$ and $C' \subseteq C$. Since C is directed by inclusion, D is a directed set, and if for $\delta = (a, C) \in D$ we define $\alpha(\delta) := a$, then $\alpha : D \to A$ is cofinal so $\{x_{\alpha(\delta)}\}_{\delta\in D}$ is a subnet of $\{x_a\}$. If $C \in \mathcal{C}$, then (since x_a is frequently in C) there is an $a_0 \in A$ such that $x_{a_0} \in C$, so $\delta_0 := (a_0, C) \in D$. If $\delta = (a, B) \in D$ and $\delta \geq \delta_0$, then $B \subseteq C$, and since $x_{\alpha(\delta)} = x_\alpha \in B$, it follows that $x_{\alpha(\delta)} \in C$, so $x_{\alpha(\delta)}$ is eventually in C. ∎

Lecture 21. Moore-Smith Convergence (Cont.)

21.1. Existence of universal subnets

At the end of the preceding lecture we proved:

21.1.1 Lemma 1. *Let $\mathcal{C}$ be a collection of subsets of a set X that is directed by inclusion. If a net $\{x_\alpha\}_{\alpha\in A}$ in X is frequently in each member of $\mathcal{C}$, then some subnet is eventually in each member of $\mathcal{C}$.*

Proof. Order $D := \{(a, C) \in A \times \mathcal{C} \mid x_a \in C\}$, by $(a, C) \leq (a', C')$ if $a \leq a'$ and $C' \subseteq C$. Since C is directed by inclusion, D is a directed set, and if for $\delta = (a, C) \in D$ we define $\alpha(\delta) := a$, then $\alpha : D \to A$ is cofinal so $\{x_{\alpha(\delta)}\}_{\delta\in D}$ is a subnet of $\{x_a\}$. If $C \in \mathcal{C}$, then (since x_a is frequently in C) there is an $a_0 \in A$ such that $x_{a_0} \in C$, so $\delta_0 := (a_0, C) \in D$. If $\delta = (a, B) \in D$ and $\delta \geq \delta_0$, then $B \subseteq C$, and since $x_{\alpha(\delta)} = x_\alpha \in B$, it follows that $x_{\alpha(\delta)} \in C$, so $x_{\alpha(\delta)}$ is eventually in C. ∎

In what follows, if $\{x_\alpha\}_{\alpha\in A}$ is a net in a set X then we will denote by $\mathcal{E}(x_\alpha)$ the collection of all subsets S of X such that x_α is eventually in S, and we will say that a collection $\mathcal{C}$ of subsets of X satisfies properties P1 and P2 (with respect to x_α) if

P1. Any finite number of subsets from $\mathcal{C}$ have non-empty intersection.

P2. x_α is frequently in each $S \in \mathcal{C}$.

We note that $\mathcal{E}(x_\alpha)$ clearly satisfies properties P1 and P2, and that if $\mathcal{C}$ is *maximal* with respect to satisfying P1 and P2, then $C_1, C_2 \in \mathcal{C} \implies C_1 \cap C_2 \in \mathcal{C}$.

21.1.2 Lemma 2. *Let $\{x_\alpha\}_{\alpha\in A}$ be a net in a set X and let $\mathcal{C}$ be a collection of subsets of X that includes $\mathcal{E}(x_\alpha)$ and is maximal with* *respect to satisfying properties P1 and P2. Then if S is any subset of X, either S or $X \setminus S$ belongs to $\mathcal{C}$.*

Proof. Assuming $S \notin \mathcal{C}$ and will derive $X \setminus S \in \mathcal{C}$ as a consequence. Since $\mathcal{C}$ is maximal with respect to satisfying properties P1 and P2 it follows that either:

i) $S \cap C = \emptyset$ for some $C \in \mathcal{C}$, or

ii) x_α is not frequently in S,

for otherwise we could append S to $\mathcal{C}$, getting a larger collection that satisfied P1 and P2, contradicting he maximality of $\mathcal{C}$.

If i) holds then $C \subseteq X \setminus S$, and since x_α is frequently in C, it is also frequently in $X \setminus S$. But since $C \cap C' \neq \emptyset$ for all $C' \in \mathcal{C}$, it also follows that $(X \setminus S) \cap C' \neq \emptyset$ for all $C' \in \mathcal{C}$, so the maximality of $\mathcal{C}$ implies that $X \setminus S \in \mathcal{C}$.

If on the other hand ii) holds then, as pointed out above, x_α is eventually in $X \setminus S$, i.e., $X \setminus S$ belongs to $\mathcal{E}(x_\alpha)$, and since $\mathcal{E}(x_\alpha) \subseteq \mathcal{C}$, $X \setminus S \in \mathcal{C}$ in this case also. ∎

21.1.3 Theorem. *Any net $\{x_\alpha\}_{\alpha \in A}$ in a set X has a subnet that is a universal net for X.*

▷ **Exercise 21–1.** Prove this theorem. Hint: Here is a strategy. Use Zorn's Lemma to show that there exists a collection $\mathcal{C}$ of subsets of X that includes $\mathcal{E}(x_\alpha)$, and is maximal with respect to satisfying P1 and P2. Check that this $\mathcal{C}$ is directed by inclusion, so that by Lemma 1, there is a subnet $\{x_{\alpha(\delta)}\}_{\delta \in D}$ that is eventually in each subset of $\mathcal{C}$. Finally, use Lemma 2 to deduce that this subnet is universal.

21.1.4 Corollary. *A space X is compact if and only if every universal net in X is convergent.*

▷ **Exercise 21–2.** Prove this corollary. Hint: We saw earlier that X is compact if and only if every net in X has a convergent subnet, and also that a universal net is convergent if and only if it has a convergent subnet.

21.2. Subbases and Weak Topologies

Definition. If X is any set and $\mathcal{T}_1$ and $\mathcal{T}_2$ are two topologies for X, then we say that $\mathcal{T}_1$ is *stronger* than $\mathcal{T}_2$ or that $\mathcal{T}_2$ is *weaker* than $\mathcal{T}_1$ if $\mathcal{T}_2 \subseteq \mathcal{T}_1$, i.e., if $\mathcal{T}_1$ has more open sets than $\mathcal{T}_2$.

Since the intersection of any family of topologies is clearly a topology, if $\mathcal{S}$ is a collection of subsets of a set X, then there is a weakest topology that includes $\mathcal{S}$, namely the intersection of all the topologies that include $\mathcal{S}$. We call this *the topology generated by* $\mathcal{S}$.

Definition. A collection $\mathcal{S}$ of open sets of a topological space X is called a *subbase* for X if $\mathcal{S}$ generates the topology of X, i.e., if the topology of X is the weakest topology that includes $\mathcal{S}$.

21.2.1 Proposition. *A collection $\mathcal{S}$ of subsets of a space X is a subbase for X if and only if X and $\emptyset$ together with all finite intersections of sets from $\mathcal{S}$ is a base for the open sets of X.*

▷ **Exercise 21–3.** Prove this.

The following corollary is an easy consequence of the fact that for any function $f : X \to Y$ the map f^{-1} from subsets of Y to subsets of X preserves all Boolean operations.

21.2.2 Corollary. *If X and Y are topological spaces, then a function $f : X \to Y$ is continuous provided that $f^{-1}(O)$ is open in X for every set O belonging to a subbase for the open sets of Y.*

21.2.3 Proposition. *If $f_\alpha : X \to Y_\alpha$ is an indexed family of functions from a set X into topological spaces Y_α, then there is a weakest topology $\mathcal{T}$ for X making all the f_α continuous. If $\mathbf{S}_\alpha$ is a subbase for the open* *sets of Y_α then the collection $\mathbf{S}$ of subsets of X the form $f_\alpha^{-1}(S_\alpha)$, where $S_\alpha \in \mathbf{S}_\alpha$, is a subbase for $\mathcal{T}$. A function $f : Z \to X$ of a space Z into X is continuous in this topology if and only if each of the functions $f_\alpha \circ f : Z \to Y_\alpha$ is continuous.*

▷ **Exercise 21–4.** Prove this proposition.

Remark. The topology $\mathcal{T}$ of the above proposition is usually referred to as the *weak topology* defined by the functions f_α.

Example. Let V be a topological vector space and V^* its dual vector space of continuous linear maps $\ell : V \to \mathbf{R}$. The *weak topology for V* is the weak topology defined by V^*, and the weak* topology for V^* is the weak topology defined by the evaluation maps, $Ev_x : V^* \to \mathbf{R}$ for all $x \in V$, where $Ev_x(\ell) := \ell(x)$.

21.2.4 Proposition. *If $f_\alpha : X \to Y_\alpha$ is an indexed family of functions from a set X into topological spaces Y_α and x_δ is a net in X, then $x_\delta \to x$ with respect to the weak topology defined by the f_α if and only if $f_\alpha(x_\delta) \to f(x)$ for each $\alpha \in A$.*

▷ **Exercise 21–5.** Prove this proposition.

As we shall see next, this "weak topology" construction is also the tool for defining the topology of an arbitrary product of topological spaces.

Lecture 22.
Product Spaces and Tychonoff's Theorem.

22.1. The Product Topology

Let $\{X_\alpha\}_{\alpha \in A}$ be an indexed family of sets. That is, A is a set (called the index set) and $\alpha \mapsto X_\alpha$ is a function from A into some collection of sets. We define the *product set* $\prod\limits_{\alpha \in A} X_\alpha$ to be the set of all functions $x : \alpha \mapsto x(\alpha)$ with domain A such that $x(\alpha) \in X_\alpha$ for each $\alpha \in A$. The usual way to think about elements of the product set is as "choice functions", since they choose an element $x(\alpha)$ from each of the sets X_α. (So one statement of the Axiom of Choice is that the product of non-empty sets is non-empty.) Note that for each $a \in A$ we have a natural "projection" map, Π_a, from the product $\prod\limits_{\alpha \in A} X_\alpha$ into X_a defined by $\Pi_a(x) := x(a)$.

In case each X_α is a topological space, we define the *product topology* for the product space $\prod\limits_{\alpha \in A} X_\alpha$ to be the weakest topology for $\prod\limits_{\alpha \in A} X_\alpha$ making all the projection maps $\{\Pi_a \mid a \in A\}$ continuous. Note that there is an important special case of the product space construction. If $\mathcal{C}$ is any collection of sets, we have a product set $\prod\limits_{S \in \mathcal{C}} S$. Here we are looking at $\mathcal{C}$ as the index set and the map $\alpha \mapsto X_\alpha$ above is just the identity map—i.e., the elements of $\mathcal{C}$ are "self-indexing". So in this case, an element of the product is a map x defined on $\mathcal{C}$ that for each $S \in \mathcal{C}$ selects an element $x(S)$ belonging to S, and the projection map Π_S maps x to $x(S)$. The advantage of the more general formulation in terms of indexed families of sets is that it allows the same set to occur more than once as an X_α. An important special case of this is the set $\mathbf{R}^{\mathbf{R}}$ of all functions $f : \mathbf{R} \to \mathbf{R}$. We can look at this as the product $\prod\limits_{x \in \mathbf{R}} X_x$ where for each $x \in \mathbf{R}$, $X_x = \mathbf{R}$. Note that the projection maps Π_x are just the evaluation maps, $\Pi_x(f) := f(x)$ so, by the proposition below, the product topology in this case is what is usually called the topology of pointwise convergence; i.e., a net f_δ converges to f in the product topology if and only if $f_\delta(x) \to f(x)$ for each $x \in \mathbf{R}$.

The following propositions are special cases of earlier more general results for the weak topology defined by a family of functions from a set to various topological spaces.

22.1.1 Proposition. *If $\{X_\alpha\}_{\alpha\in A}$ is an indexed family of topological spaces then a net x_δ in $\prod\limits_{\alpha\in A} X_\alpha$ converges to x in the product topology if and only if $x_\delta(a) \to x(a)$ for each $a \in A$.*

22.1.2 Proposition. *If $\{X_\alpha\}_{\alpha\in A}$ is an indexed family of topological spaces, then a function $F : Z \to \prod\limits_{\alpha\in A} X_\alpha$ mapping a space Z into their product is continuous in the product topology if and only if, for each projection map Π_a, the composition $\Pi_a \circ F : Z \to X_a$ is continuous.*

22.1.3 Proposition. *A subbase for the product topology for $\prod\limits_{\alpha\in A} X_\alpha$ consists sets of the sets $\Pi_a^{-1}(U)$ where $a \in A$ and U is open in X_a. Such a subset of the product is called a cylinder set over X_a and has the form $\prod\limits_{\alpha\in A} O_\alpha$ where $O_a = U$ and $O_\alpha = X_\alpha$ for $\alpha \neq a$.*

Remark. A finite intersection of cylinder sets has the form $\prod\limits_{\alpha\in A} O_\alpha$, where each O_α is open in X_α and $O_\alpha = X_\alpha$ except for finitely many $\alpha \in A$. Hence sets of this form are a basis for the product topology.

22.1.4 Tychonoff's Theorem. *The product $\prod\limits_{\alpha\in A} X_\alpha$ of non-empty spaces is compact if and only if each factor X_α is compact.*

Proof. If every factor is non-empty, each projection map is surjective, so if the product is compact then each factor is the continuous image of a compact space and hence is compact.

Conversely suppose each factor is compact and let $\{x_\delta\}_{\delta\in D}$ be a universal net in the product. Then $x_\delta(a) = \Pi_a(x_\delta)$ is a universal net in X_a for any $a \in A$ and, since X_a is compact, it converges to some

$x(a) \in X_a$. This defines an element $x \in \prod_{\alpha \in A} X_\alpha$ such that $x_\delta \to x$, so we have verified that every universal net in the product converges and hence that the product is compact. ∎

22.1.5 Banach-Alaoglu Theorem. *If V is a Banach space then the closed unit ball in its dual space is compact in the weak* topology.*

Proof. For each $v \in V$ let $D_v := \{r \in \mathbf{R} \mid |r| \leq \|v\|\}$. Since each D_v is compact, so is their product $D := \prod_{v \in V} D_v$, and it will suffice to show that B, the closed unit ball in V^* is a closed subspace of D. Now an element of D is an arbitrary function $f : V \to \mathbf{R}$ such that $f(v) \leq \|v\|$. Recalling that the norm for V^* is defined by $\|\ell\| := \sup\{|\ell(v)|/\|v\| \mid v \in V\}$ it follows that an element of B is just a linear map $\ell : V \to \mathbf{R}$ satisfying $\ell(v) \leq \|v\|$, so in fact $B \subseteq D$. Moreover, since the evaluation map $v \mapsto f(v)$ is just the projection of D on D_v, the weak* for B is just its topology as a subspace of D, and the only thing left to verify is that B is closed in D. But this just says that a pointwise limit of linear functions is linear, which is obvious. ∎

Lecture 23.
Connected Spaces and Sets

23.1. Connectedness

Two subsets A and B of a space X are called *separated* (in X) if $\overline{A} \cap B = A \cap \overline{B} = \emptyset$; i.e., if no point of B is a limit point of A and vice versa. We note that if in addition $A \cup B = X$, then

$$\overline{A} = \overline{A} \cap (A \cup B) = (\overline{A} \cap A) \cup (\overline{A} \cap B) = A,$$

and similarly $\overline{B} = B$. In other words, if X is the union of two separated sets, then each of them is closed in X, and since each is the complement of the other, each of them is also open in X. Conversely of course if X is the union of two complementary sets each of which is open and closed in X, then these sets are separated.

Definition. A topological space X is called *connected* if the only two subsets of X that are both open and closed are X itself and the empty set $\emptyset$. A subset of X is called connected if it is connected in the subspace topology.

Let us denote by $\mathbf{Z}_2$ the two-point set $\{0, 1\}$ with the discrete topology.

23.1.1 Theorem. *If X is a topological space then the following are equivalent.*

1) *X is not the union of two non-empty separated sets.*
2) *X is not the union of two non-empty, disjoint, closed sets.*
3) *X is not the union of two non-empty, disjoint, open sets.*
4) *X is connected.*
5) *Any continuous map $f : X \to D$ into a discrete space D is constant.*
6) *Any continuous map $f : X \to \mathbf{Z}_2$ is constant.*

Proof. We saw the equivalence of 1), 2) and 3) above. If 3) holds and $G \subseteq X$ is both open and closed in X, then $\complement G$ is also open in X, so by 3) one of G and $\complement G$ is empty and the other is all of X. This shows that X is connected, so 3) $\implies$ 4). If 4) holds and f is a map into a discrete space D with $f(x) = d$, then since $\{d\}$ is open and closed in D, $f^{-1}(d)$ is open and closed in X. Then, since X is connected and $x \in f^{-1}(d)$, it follows that $f^{-1}(d) = X$. Thus, f is constant, so 4) implies 5). That 5) $\implies$ 6) is trivial. Finally if A and B are disjoint open subsets of X then we can define a continuous function $f : X \to \mathbf{Z}_2$ that equals 0 in A and equals 1 in B, so if 6) holds then one of A and B is empty and the other is all of X, showing that 6) $\implies$ 3). ∎

23.1.2 Proposition. *The continuous image of a connected space is connected.*

▷ **Exercise 23–1.** Prove this. Hint: Suppose X is a connected space and that $f : X \to Y$ is a continuous, surjective map. If $g : Y \to \mathbf{Z}_2$ be a continuous map, then $g \circ f : X \to \mathbf{Z}_2$ is continuous.

23.1.3 Proposition. *If A is a connected subset of a space X and $A \subseteq B \subseteq \overline{A}$ then B is also connected. In particular, the closure of a connected set is conected.*

▷ **Exercise 23–2.** Prove this. Hint: If $g : B \to \mathbf{Z}_2$ is continuous, then the restriction of g to A is constant.

23.1.4 Proposition. *Let $\mathcal{C}$ be a collection of connected subsets of a space X. If the intersection of every pair of elements of $\mathcal{C}$ is non empty, then the union of all the sets in $\mathcal{C}$ is a connected subset of X.*

▷ **Exercise 23–3.** Prove this. Hint: If Y is the union of all the sets in $\mathcal{C}$, assume $g : Y \to \mathbf{Z}_2$ is continuous and show it must be constant.

23.1.5 Proposition and Definition. *Every point x of a space X is contained in a unique maximal connected subset C_x of X called the* **connected component** *of x, and these connected components partition X into disjoint, closed subsets.*

Proof. By the preceding proposition, the union of all the connected subsets of X that contain a point x is a connected set C_x, and it is clearly the largest connected subset of X that contains x. Since the closure of a connected set is connected, it follows that each C_x is closed. If C_x and C_y have a common point then their union would be connected and by maximality they must be equal. ∎

23.2. The Connected Subsets of R

If the connected component of each point of a space is just the point itself the space is called *totally disconnected*. We have not yet shown the existence of any interesting connected sets, and as far as we know at this point all spaces might be totally disconnected—in which case connectedness would be a pretty useless concept. As I am sure you know, quite the opposite is the case, and to see some of the utility of connectedness we consider next the question of identifying the connected subsets of **R**.

Let us say that a subset S of **R** has the *between-ness property* if $a, b \in S$ and $a < c < b$ implies that $c \in S$. As is pretty obvious, and as the following exercise make explicit, the subsets having the between-ness property are just what are usually called "intervals", and we shall use the term interval as a synonym for a set with the between-ness property.

▷ **Exercise 23–4.** By considering the following cases:

a) S is (or is not) bounded above (below),

b) S does (or does not) contain its least upper (greatest lower) bound,

show that if S is a subset of **R** that satisfies the between-ness property then it has one of the following forms:

i) (a, b) where $-\infty \le a \le b \le \infty$,
ii) $[a, b)$ where $-\infty < a < b \le \infty$,
iii) $(a, b]$ where $-\infty \le a \le b < \infty$.
iv) $[a, b]$ where $-\infty < a \le b < \infty$.

23.2.1 Lemma. *Let $S \subseteq \mathbf{R}$ be an interval and let $f : S \to \mathbf{Z}_2$ be continuous. If $x_0 \in S$ then $f(x) = f(x_0)$ for all $x \in S$ with $x \ge x_0$.*

Proof. We note that since $\mathbf{Z}_2$ is discrete, f is locally constant. Let $A = \{x \in S \mid f \text{ is constant on } [x_0, x]\}$. If the lemma is false then A is bounded above, and we let M be its least upper bound. Clearly $[x_0, M) \subseteq S$ and by continuity, f is constant on $[x_0, M)$. If $M \in S$ then, again by continuity, f is constant on $[0, M]$. But then M must be the right-hand endpoint of S, proving the lemma, for otherwise M is interior to S and since f is locally constant, f would be constant on $[0, M + \epsilon)$ for some $\epsilon > 0$, which contradicts the choice of M. ∎

23.2.2 Theorem. *A subset of $\mathbf{R}$ is connected if and only if it is an interval.*

▷ **Exercise 23–5.** Prove this theorem. Hint: Suppose $S \subseteq \mathbf{R}$ is not an interval, so there exist a, b, c with $a < c < b$, $a, b \in S$ and $c \notin S$. Show that $A := \{x \in S \mid x < c\}$ and $B := \{x \in S \mid x > c\}$ is a decomposition of S into two separated sets, so S is not connected. For the converse, recall that it suffices to show that any continuous $f : S \to \mathbf{Z}_2$ is constant, and that follows easily from the lemma.

23.2.3 Corollary—Intermediate Value Theorem. *If a continuous real-valued function on a connected space takes on two values it also takes on all intermediate values. In particular if a continuous real-valued function on an interval has both positive and negative values, it has a zero somewhere in the interval.*

23.2.4 Corollary. *Every real polynomial of odd degree has a root.*

▹ **Exercise 23–6.** Let Ω be a linearly ordered set made into a topological space by using the so-called "order topology", where the neighborhoods of a point ω_0 are open intervals

$$(\omega_1, \omega_2) := \{\omega \in \Omega \mid \omega_1 < \omega < \omega_2\}$$

that contain ω_0. Show that if Ω satisfies the least upper bound principle then intervals in Ω are connected.

Lecture 24.
Connected Spaces and Sets (Cont.)

24.1. Local Connectedness

Definition. A space X is called *locally connected* if each neighborhood of any point p includes a connected neighborhood of p.

24.1.1 Proposition. *A space X is locally connected if and only if the connected components of open sets are open.*

▷ **Exercise 24–1.** Prove this.

24.1.2 Proposition. *The product $X \times Y$ of two non-empty spaces is connected if and only if both X and Y are connected.*

Proof. Since X and Y are the images of $X \times Y$ under projection, one direction is clear. If both X and Y are connected then for any $x_0 \in X$ and $y_0 \in Y$, the set $X \times \{y_0\}$ and $\{x_0\} \times Y$ are connected, and since they have the common point (x_0, y_0), their union $(X \times \{y_0\}) \cup (\{x_0\} \times Y)$ is also connected. Moreover, if we keep x_0 fixed and vary y_0, all these sets sets include $\{x_0\} \times Y$, so their union $X \times Y = \bigcup_{y \in Y} (X \times \{y\}) \cup (\{x_0\} \times Y)$ is connected. ∎

Remark. The above proof becomes much more transparent and motivated if you draw a picture of the sets involved for the case $X = Y = \mathbf{R}$ with $x_0 = y_0 = 0$.

24.1.3 Corollary. $\mathbf{R}^n$ *is connected and locally connected.*

24.1.4 Corollary. *Every open subset of* $\mathbf{R}$ *is a countable disjoint union of open intervals.*

Proof. Since $\mathbf{R}$ is locally connected, components of open sets are open connected sets, hence open intervals. To check countability select a rational in each component. ∎

24.1.5 Proposition. *If I is an interval and $f : I \to \mathbf{R}$ is continuous and injective, then f is strictly monotonic and is a homeomorphism of I onto $f(I)$—i.e., f^{-1} is continuous.*

Proof. Let $a, b \in I$ with $a < b$ and assume $f(a) < f(b)$. (A similar argument works if $f(a) > f(b)$.) We will show that for all $t \in (a, b]$, $g(t) := f(t) - f(a) > 0$, so that $f(a) < f(t)$. In fact, $g(t) \neq 0$ since f is injective, and since $g(b) > 0$, by the Intermediate Value Theorem g cannot be negative anywhere. It now follows that f maps any open interval to an open interval, hence it is an open map, which means that f^{-1} is continuous. ∎

24.2. Paths and Path Connectedness

Definition. A *path* in a space X is a map $\sigma : [a, b] \to X$ of a non-degenerate closed interval in $\mathbf{R}$ into X. We call $[a, b]$ the parameter interval, $\sigma(a)$ is called the initial point of the path, $\sigma(b)$ the final point, and both are called endpoints. We also say that σ is a path in X from $\sigma(a)$ to $\sigma(b)$.

Non-degenerate means that a is strictly less than b. If $[c, d]$ is a second non-degenerate closed interval, then by a translation and a stretching, we get a homeomorphism $\phi : [c, d] \to [a, b]$; namely $\phi(t) := a + \rho(t - c)$, where $\rho := \frac{b-a}{d-c}$. Then $\sigma \circ \phi$ gives a "reparametrization" of σ with $[c, d]$ as parameter interval. In particular, we can always reparametrize with $[0, 1]$ as the parameter interval. For many purposes the original path and a reparametrized path are "equivalent".

It is important to be able to manipulate paths at both the intuitive and rigorous level. In particular the concepts of reparametrization of paths, reversing the direction of a path, and "adding" paths are essential tools. The following exercise is meant to help you check your understanding of these concepts.

▷ **Exercise 24–2.**

1) Make sure you understand the above paragraph concerning reparametrization of paths..
2) Restate the following more carefully and in more detail. Reflecting the interval $[a, b]$ in its midpoint gives a homeomorphism of $[a, b]$ onto itself, and composing a path $\sigma : [a, b] \to X$ with this map gives a path $\tilde{\sigma}$ that traces out σ in the reverse direction, in particular interchanging the initial and final endpoints.
3) Similarly, give a more detailed statement of the following. Given two paths σ_1, σ_2 such that the final point of σ_1 is the initial point of σ_2, there is a combined path $\sigma_1 * \sigma_2$ that first traces out σ_1, and then continues to trace out σ_2.

The following proposition is an immediate consequence of this exercise.

24.2.1 Proposition and Definition. *If X is a topological space then "there exists a path in X from p to q" defines an equivalence relation for X. The equivalence classes of this relation are called the* **path components** *of X, and X is called pathwise connected if it has only one path component.*

Definition. A space X is said to be *locally pathwise connected* if every point of X has a neighborhood basis consisting of pathwise connected neighborhoods.

It is immediate from its definition that the path component of a point p in a space X is the union of the images of all paths in X that have p as their initial point. Since these images are connected sets that contain p, they are included in the connected component of p. Hence:

24.2.2 Proposition. *The path component of a point p in a space X is a subset of the connected component of p in X. In particular, a pathwise connected space is connected and a locally pathwise connected space is locally connected.*

24.2.3 Proposition. *A space is locally pathwise connected if and only if the path components of open sets are open.*

24.2.4 Corollary. *In a locally pathwise connected space the path components of an open set coincide with its connected components.*

▷ **Exercise 24–3.** Prove this.

Note that a convex set in a topological vector space is clearly pathwise connected, A topological vector space is called *locally convex* if the origin has a neighborhood basis consisting of convex sets. Since the translate of a convex set again convex it follows that:

24.2.5 Proposition. *Locally convex topological vector spaces, and in particular normed spaces, are locally pathwise connected.*

Lecture 25.
Homotopy Theory

25.1. The Exponential Law

In what follows X, A, and B are topological spaces and $I = [0,1]$. We denote the continuous maps from A into X by $C(A,X)$ and also by X^A.

Definition. A continuous maps $F : A \times I \to X$ is called a *homotopy* of maps of A into X. If f_0 and f_1 are continuous maps of A into B such that $f_i(x) := F(x,i)$ then we say that F is a homotopy of f_0 with f_1. If a homotopy of f_0 with f_1 exists then we say that f_0 and f_1 are homotopic.

It is natural to conjecture that a homotopy should be the same as a path in the space of continuous maps of A into X. In case A and X are metric spaces with A compact we have defined a metric ρ_∞ on $C(A,X) = X^A$. Let's see if we can verify this conjecture.

$\triangleright$ **Exercise 25–1.** Let A, B, and X be metric spaces with A and B compact. Given $F \in X^{A\times B}$ define $\tilde{F} : B \to X^A$ by $\tilde{F}(b)(a) := F(a,b)$.

1) Show that $\tilde{F}$ is continuous, and hence $\tilde{F} \in (X^A)^B$.
2) Show that any G in $(X^A)^B$ has the form $\tilde{F}$ for a unique $F \in X^{A\times B}$.
3) Deduce the so-called "Exponential Law": $F \mapsto \tilde{F}$ is an isometry of $X^{A\times B}$ with $(X^A)^B$.

25.1.1 Proposition. *If A and X are metric spaces with A compact and f_0, f_1 are continuous maps of A to X, then a homotopy of f_0 with f_1 is the same as a path in $C(A,X)$ joining f_0 with f_1. Hence f_0 and f_1 are homotopic if and only if they are in the same path component of $C(A,X)$.*

Proof. This follows from the Exponential Law, taking $B = I$. ∎

25.2. Based Loops and the Fundamental Group

Definition. If X is a space and $x_0 \in X$, we define the space $\Lambda(X, x_0)$ of *loops in X based at x_0* to be the subspace of $C(I, X)$ consisting of all paths $\sigma : I \to X$ with initial point and final point both equal to x_0. A homotopy of loops based at x_0 is a path in $\Lambda(X, x_0)$. We denote the set of path components (called homotopy classes) of $\Lambda(X, x_0)$ by $\Pi_1(X, x_0)$, and if $\sigma \in \Lambda(X, x_0)$ then $[\sigma] \in \Pi_1(X, x_0)$ will denote its homotopy class. If $\sigma \in \Lambda(X, x_0)$ we define its *reverse* $\tilde{\sigma}$ by $\tilde{\sigma}(t) := \sigma(1-t)$. If $\sigma_1, \sigma_2 \in \Lambda(X, x_0)$, we define their *product* $\sigma_1 * \sigma_2$ by $\sigma_1 * \sigma_2(t)$ equals $\sigma_1(2t)$ if $0 \le t \le \frac{1}{2}$ and equals $\sigma_2(2t-1)$ if $\frac{1}{2} \le t \le 1$. Finally, we denote by e_{x_0} the constant loop based at x_0 and we denote its homotopy class $[e_{x_0}]$ by $\mathbf{e}_{x_0}$.

▷ **Exercise 25–2.** Show that if $\sigma_1, \sigma_2, \sigma_3, \sigma_4 \in \Lambda(X, x_0)$ then:

1) If $[\sigma_1] = [\sigma_2]$ then $[\tilde{\sigma}_1] = [\tilde{\sigma}_2]$.
2) If $[\sigma_1] = [\sigma_3]$ and $[\sigma_2] = [\sigma_4]$ then $[\sigma_1 * \sigma_2] = [\sigma_3 * \sigma_4]$
3) For any $\sigma \in \Lambda(X, x_0)$, $[\sigma * \tilde{\sigma}] = \mathbf{e}$.
4) $[(\sigma_1 * \sigma_2) * \sigma_3] = [\sigma_1 * (\sigma_2 * \sigma_3)]$

Remark. This above exercise is somewhat more "difficult" than most, not because it takes a great deal of ingenuity, but rather because it involves a lot of steps. In any case it is well worth your effort to get familiar with the concepts and the computations involved. Moreover, the exercise is mathematically important, since it has as an almost immediate consequence the following theorem stating the existence of the fundamental group of a space.

25.2.1 Theorem. *If X is a space then the set $\Pi_1(X, x_0)$ of homotopy classes of loops based at x_0 becomes a group, called the fundamental group of X, under the product defined by $[\sigma_1] * [\sigma_2] = [\sigma_1 * \sigma_2]$. The identity element is $\mathbf{e}_{x_0}$, and the inverse of $[\sigma]$ is $[\tilde{\sigma}]$.*

Remark. The fundamental group $\Pi_1(X, x_0)$ seems to depend on the base point x_0 as well as on the space X, and indeed it does. However if x_0 and x_1 are in the same path component then $\Pi_1(X, x_0)$ and $\Pi_1(X, x_1)$ are isomorphic, so if X is pathwise connected then the fundamental group is well-defined up to isomorphism. Given a path γ joining x_1 to x_0, define a map $\sigma \mapsto \sigma_\gamma$ of $\Lambda(X, x_0)$ into $\Lambda(X, x_1)$ by defining σ_γ to be γ followed by σ followed by the reverse of γ.

$\triangleright$ **Exercise 25–3.** Verify that this induces an isomorphism of $\Pi_1(X, x_0)$ with $\Pi_1(X, x_1)$

Remark. So when x_0 and x_1 are in the same path component, $\Pi_1(X, x_0)$ and $\Pi_1(X, x_1)$ are isomorphic, however they are **not** canonically isomorphic. That is, the isomorphism above depend on the homotopy class of the path γ joining x_0 to x_1.

$\triangleright$ **Exercise 25–4.** If $f : X \to Y$ is continuous and $\sigma \in \Lambda(X, x_0)$ then $f_*(\sigma) := f \circ \sigma \in \Lambda(Y, f(x_0))$. Show that this map

$$f_* : \Lambda(X, x_0) \to \Lambda(Y, f(x_0))$$

induces a group homomorphism $\Pi_1(f) : \Pi_1(X, x_0) \to \Pi_1(Y, f(x_0))$ and show that if $f_0 : X \to Y$ and $f_1 : X \to Y$ are homotopic via a homotopy that keeps the image of x_0 fixed, then $\Pi_1(f_0) = \Pi_1(f_1)$.

Fiber Bundles and Covering Spaces. The final topic mentioned in the course syllabus is “Covering Spaces”. It so happens that there is a much more general concept called a fiber bundle, and while it is true that covering spaces are simpler objects that can be and often are considered on their own terms, I find it more intuitive and better motivated to consider a covering space as a special kind of fiber bundle. In fact, what we shall see is that a covering space is “just a fiber bundle with discrete fiber”. There is also another excellent reason to introduce fiber bundles at this point; they turn out to play a very important role in the second and third quarters of this course.

Lecture 26. Fiber Bundles and Covering Spaces

26.1. Mapping Triples

By a *mapping triple* we will mean an ordered triple (E, B, π), where E and B are locally pathwise connected spaces called the *total space* and *base space* respectively, and $\pi : E \to B$ is a continuous surjection called the *projection*. For $p \in B$ we call $F_p = \pi^{-1}(p)$ the *fiber* over p. If (E_1, B, π_1) and (E_2, B, π_2) are two mapping triples with the same base space, then a homeomorphism $\Phi : E_1 \to E_2$ is called an *equivalence of mapping triples* if $\pi_2 \circ \Phi = \pi_1$. We note that this says that Φ maps the fiber of (E_1, B, π_1) over any $b \in B$ homeomorphically onto the fiber of (E_2, B, π_2) over b.

If B and F are spaces, and Π is the usual projection of $B \times F$ onto B, then the mapping triple $(B \times F, B, \Pi)$ is called the *product fiber bundle over B with fiber F*, and we note that the its fiber over every point of B can be identified with F. A mapping triple (E, B, π) is called a *trivial fiber bundle with fiber F* if it is equivalent to the product fiber bundle $(B \times F, B, \Pi)$.

Definition. The mapping triple (E, B, π) is called a *locally trivial fiber bundle with fiber F* if each point $b \in B$ has an open neighborhood O such that mapping triple $(\pi^{-1}(O), O, \pi)$ is a trivial fiber bundle with fiber F, and such an O is called a trivializing neighborhood for (E, B, π) and an equivalence $\Phi : O \times F \to \pi^{-1}(O)$ is called a trivialization of (E, B, π) over O.

Definition. If (E, B, π) is a mapping triple and O is open in B, then a continuous map $\chi : O \to E$ is called a *local cross-section* for (E, B, π) (over O) if $\pi \circ \chi$ is the identity map of O, i.e., if for each $b \in O$, χ "chooses" an element of the fiber over b. (If $\Phi : O \times F \to \pi^{-1}(O)$ is a trivialization of (E, B, π) over O, note that each $f \in F$, gives is a local cross-section $b \mapsto \Phi(b, f)$ over O.)

Example. The Möbius Strip is a locally trivial (but not a trivial) fiber bundle over the circle with fiber a closed interval.

Definition. Let(E, B, π) be a mapping triple. An open set O in the base B is said to be *evenly covered* if $\pi^{-1}(O)$ is the disjoint union of open sets of E each of which is mapped homeomorphically onto O by π. A *covering space* is a mapping triple such that every point of the base has an evenly covered open neighborhood.

▷ **Exercise 26–1.** Show that a locally trivial fiber bundle with a discrete fiber is a covering space. Conversely, prove that if (E, B, π) is a covering space and O is an open set in B that is evenly covered, then $(\pi^{-1}(O), O, \pi)$ is a trivial fiber bundle with discrete fiber, and deduce that if B is connected then (E, B, π) is a locally trivial fiber bundle with discrete fiber. (Hint: Show that the set of points p in B such that the fiber at p has some fixed cardinality is open.)

The remainder of this section is a discussion of a number of examples to help you familiarize yourself with the concepts of a fiber bundle and a covering space.

Example. If we regard $\mathbf{S}^1$ as the unit circle in the complex plane and define $\pi : \mathbf{R} \to \mathbf{S}^1$ by $\pi(x) := e^{2\pi i x}$, then the mapping triple $(\mathbf{R}, \mathbf{S}^1, \pi)$ becomes a covering space with fiber $\mathbf{Z}$, the integers.

Example. If we define $\pi : \mathbf{S}^1 \to \mathbf{S}^1$ by $\pi(z) := z^n$, then $(\mathbf{S}^1, \mathbf{S}^1, \pi)$ becomes a covering space whose fiber at any point z_0 in the base is the set of n-th roots of z_0.

Example. The boundary of the Möbius Strip is a covering space over the circle with fiber $\mathbf{Z}_2$.

Quotient Spaces. If $\sim$ is an equivalence relation on a set X then we denote the equivalence class of $x \in X$ by $[x]$, denote the quotient set of all equivalence classes by $X/\sim$, and denote by π the quotient or projection map $x \mapsto [x]$ of X onto $X/\!\sim$. If X is a topological space the natural topology for $X/\!\sim$, called the *quotient topology*; is the strongest topology making the projection continuous, so that a subset O of $X/\!\sim$ is open if and only if its inverse image $\pi^{-1}(O)$ (the union of all the equivalence classes contained in O) is open in X. It is easy to see that a map f of $X/\!\sim$ to a space Y is continuous if and only if $f \circ \pi : X \to Y$ is continuous.

Example. The quotient space obtained from the n-dimensional sphere, $\mathbf{S}^n$ by identifying pairs of antipodal points is called the real projective space of dimension n, $\mathbf{RP}^n$. Since each line through the origin in $\mathbf{R}^{n+1}$ intersects $\mathbf{S}^n$ in a pair of antipodal points, $\mathbf{RP}^n$ can also be identified with the space of all one-dimensional subspaces of $\mathbf{R}^{n+1}$. The natural projection of $\mathbf{S}^n$ onto $\mathbf{RP}^n$ is clearly a "two-sheeted" covering space—i.e., its fiber is $\mathbf{Z}_2$.

Remark. Recall that if G is a group and H is a subgroup, then the equivalence relation on G defined by $g_1 \sim g_2$ if and only if $g_1^{-1} g_2 \in H$ has as equivalence classes the left cosets of H. If G is a topological group (meaning G has a topology for which group multiplication and $g \mapsto g^{-1}$ are continuous) and H is closed in G, then $H \times H$ is closed in $G \times G$, so the space G/H of left cosets with the quotient topology is a Hausdorff space. If $\pi : G \to G/H$ is the usual coset map $g \mapsto gH$ then we get an important mapping triple $(G, G/H, \pi)$ with H as fiber, and it is natural to ask when this is a locally trivial fiber bundle. The following two exercises provide the answer.

$\triangleright$ **Exercise 26–2.** Show that if H is a discrete subgroup of a G then $(G, G/H, \pi)$ is a covering space, and show that if we identify $\mathbf{R}/\mathbf{Z}$ with $\mathbf{S}^1$ in the usual way then the covering space $(\mathbf{R}, \mathbf{R}/\mathbf{Z}, \pi)$ is equivalent to the example $(\mathbf{R}, \mathbf{S}^1, \pi)$ above.

$\triangleright$ **Exercise 26–3.** More generally, show $(G, G/H, \pi)$ is a locally trivial fiber bundle if and only if there exists a local cross-section $\chi_e : O \to G$ in some open neighborhood of the identity, e. (Hint: First show that given such a χ_e we can get a local cross-section χ_g defined in the open neighborhood gO of g by "conjugating χ by g", then use χ_g to define in a natural way a map $\Phi : gO \times H \to G$ and show that it trivializes $(G, G/H, \pi)$ over gO.)

Lecture 27. Liftings and the Covering Homotopy Theorem

27.1. Liftings

In what follows we assume all spaces are locally pathwise connected. We denote the unit interval $[0, 1]$ by I, and $\partial I := \{0, 1\}$ denotes its boundary.

Definition. Let (E, B, π) be a mapping triple and $f : X \to B$ a continuous map of a space X into the base space. A *lifting* of f is a continuous map $F : X \to E$ of X into the total space such that $f = \pi \circ F$.

Remark. We have already considered one special kind of lifting; a local cross-section over an open subset O of B is just a lifting of the identity map of O. Another important special case is a so-called a deck transformation of a covering space.

Definition. An equivalence $\Phi : E \to E$ of a covering space (E, B, π) with itself is called a *deck transformation* of the covering space. These clearly form a group under composition, called the group of deck transformations of (E, B, π).

Remark. By definition of an equivalence, if $\Phi : E \to E$ is a deck transformation of (E, B, π), then Φ is a homeomorphism of E with itself such that $\pi \circ \Phi = \pi$. Thus by definition of a lifting, Φ is a lifting of π. It follows by the Uniqueness of Liftings corollary, proved below, that two deck transformations that agree at one point are in fact equal.

$\triangleright$ **Exercise 27–1.** For the example $(\mathbf{R}, \mathbf{S}^1, \pi)$ of a covering space considered earlier, where $\pi(t) := e^{2\pi i t}$, show that the group of deck transformations is the subgroup $\mathbf{Z}$ of $\mathbf{R}$ acting by translation. Similarly, for the example $(\mathbf{S}^1, \mathbf{S}^1, \pi)$ with $\pi(z) := z^n$, show that the group of deck transformations is the subgroup Γ of $\mathbf{S}^1$ consisting of the n-th roots of unity, acting by translation (i.e., rotation).

27.1.1 Path Lifting Theorem. *Let (E, B, π) be a covering space, $\sigma : I \to B$ a path in B with initial point $b = \sigma(0)$, and $e \in E$ any point in the fiber $\pi^{-1}(b)$. Then there is a unique lifting $\tilde{\sigma}$ of σ such that $\tilde{\sigma}(0) = e$, and if σ is a constant path then so is $\tilde{\sigma}$. Moreover $\tilde{\sigma}$ is a continuous function of σ, i.e., if σ_n is a sequence of paths in B that converge uniformly to σ, and if $\tilde{\sigma}_n$ are liftings such that $e_n = \tilde{\sigma}_n(0)$ converges to $e = \tilde{\sigma}(0)$, then $\tilde{\sigma}_n$ converges uniformly to σ.*

Proof. First consider the special case that the image of σ lies in a single evenly covered open set O of B. Then $\pi^{-1}(O)$ is the disjoint union of open sets O_α such that the restriction π_α of π to O_α is a homeomorphism onto O. If $e \in O_a$, then since the image of a lifting is connected it must lie in a single O_α, hence $\tilde{\sigma} = \pi_a^{-1} \circ \sigma$ is clearly the unique lifting with initial point e. Moreover eventually $e_n \in O_a$, hence $\tilde{\sigma}_n = \pi_a^{-1} \circ \sigma_n$, and it follows that $\tilde{\sigma}_n$ converges uniformly to $\tilde{\sigma}$.

In the general case, the compact interval I is covered by open sets $\sigma^{-1}(O_\alpha)$, where the O_α are evenly covered. If we choose a positive integer n so large that $\frac{1}{n}$ is a Lebesgue number for this cover, then we can apply the special case argument to the restriction of σ to each subinterval $[\frac{j}{n}, \frac{j+1}{n}]$, $j = 0, \dots n-1$, and the general case follows by an easy induction. The fact that the lift of a constant path is a constant path follows from uniqueness. ∎

27.1.2 Corollary (Uniqueness of Liftings). *If X is a pathwise connected space, $f : X \to B$ a continuous map, and F_1, F_2 are two liftings of f that agree at one point, then $F_1 = F_2$.*

Proof. Assume $F_1(x) = F_2(x) = e$ for some $x \in X$. Given $x' \in X$, since X is pathwise connected there is a path γ in X with initial point x and final point x'. If $\sigma := f \circ \gamma : I \to B$, then since $\pi \circ F_1 = \pi \circ F_2 = f$, $F_1 \circ \gamma$ and $F_2 \circ \gamma$ are two liftings of σ with the same initial point e, so by the Theorem they are equal, and in particular their endpoints are equal, i.e., $F_1(x') = F_2(x')$. ∎

Definition. If $f_0, f_1 : A \to X$ are continuous maps, a homotopy F of f_0 with f_1 is said to be *fixed on a subset* S of A if $t \mapsto F(s,t)$ is a constant path for all $s \in S$. If such a homotopy F exists we say that f_0 and f_1 are *homotopic relative to* S. (Note that it follows that f_0 and f_1 agree at each point of S.) In the special case that $A = I = [0,1]$ and $S = \partial I = \{0,1\}$, if σ_0 and σ_1 are two paths in X with the same endpoints x_0 and x_1 and if they are homotopic by a homotopy fixed on ∂I then we say that σ_0 and σ_1 are *homotopic with fixed endpoints*.

We shall prove the following very important fact in the next lecture.

The Covering Homotopy Theorem. Let (E, B, π) be a covering space, $f : X \to B$ a continuous map of a space X into the base space, and $h : X \times I \to B$ a homotopy of f. If there is a lifting $F : X \to E$ of f, then there is a unique lifting $H : X \times I \to E$ of h to a homotopy of F. Moreover if the homotopy h is fixed on a subset A of X then its lifting H is also fixed on A.

Lecture 28.
Applications of The Covering Homotopy Theorem

28.1. The Covering Homotopy Theorem

28.1.1 The Covering Homotopy Theorem. *Let (E, B, π) be a covering space, $f : X \to B$ a continuous map of a space X into the base space, and $h : X \times I \to B$ a homotopy of f. If there is a lifting $F : X \to E$ of f, then there is a unique lifting $H : X \times I \to E$ of h to a homotopy of F. Moreover if the homotopy h is fixed on a subset A of X then its lifting H is also fixed on A.*

Proof. Note that if X is a point then this reduces to the Path Lifting Theorem, since a homotopy of a map of a point is the same thing as a path. In fact we can use the Path Lifting Theorem to prove the more general Covering Homotopy Theorem. For $x \in X$, let $t \mapsto H(x,t)$ be the unique path in E that starts at $F(x)$ and covers the path $t \mapsto h(x,t)$. The only thing left to prove is that this map H is continuous. Assume a sequence x_n converges to x in X. Then the subset Y of X consisting of the x_n and x is compact, so h is uniformly continuous on $Y \times I$, and hence the sequence of paths $t \mapsto h(x_n,t)$ converges uniformly to the path $t \mapsto h(x,t)$. But then by the Path Lifting Theorem, $t \mapsto H(x_n,t)$ converges uniformly to $t \mapsto H(x,t)$, proving that H is continuous. ∎

28.1.2 Corollary. *Let (E, B, π) be a covering space and let σ_0 and σ_1 are two paths in B with the same endpoints that are homotopic with fixed endpoints. If $\tilde{\sigma}_0$ and $\tilde{\sigma}_1$ are liftings of σ_0 and σ_1 respectively with the same initial point, then $\tilde{\sigma}_0$ and $\tilde{\sigma}_1$ are also homotopic with fixed endpoints, and in particular $\tilde{\sigma}_0(1) = \tilde{\sigma}_1(1)$.*

28.1.3 Corollary. *Let (E, B, π) be a covering space and let σ be a loop in B that is homotopic in B with fixed endpoints to a constant* *loop. Then any lift of σ is a loop in E that is homotopic in E with fixed endpoints to a constant loop.*

28.1.4 Corollary. *Let (E, B, π) be a covering space, $e_0 \in E$ and $b_0 = \pi(e_0) \in B$. Then $\Pi_1(\pi) : \Pi_1(E, e_0) \to \Pi_1((B, b_0)$, the induced map of fundamental groups, is injective with image the homotopy classes of loops in B based at b_0 that lift to loops based at e_0.*

▷ **Exercise 28–1.** The proofs of all the above corollaries are straightforward, but you should carry out the details to make sure that you understand all the new concepts.

28.1.5 Theorem. *If a pathwise connected space B has a non-trivial covering space (E, B, π) with a pathwise connected total space E, then $\Pi_1(B, b_0)$ is not the trivial group.*

Proof. Let γ be a path in E joining two distinct points of $\pi^{-1}(b_0)$. Then its projection $\sigma := \pi \circ \gamma \in \Lambda(B, b_0)$ is a loop in B whose lift to a path in E is not a loop in E, hence its homotopy class $[\sigma]$ is not in the image of $\Pi_1(\pi) : \Pi_1(E, e_0) \to \Pi_1(B, b_0)$ and in particular it is not the identity element of $\Pi_1(B, b_0)$. ∎

28.1.6 Corollary. *$\Pi_1(\mathbf{S}^1, 1)$ is non-trivial.*

▷ **Exercise 28–2.** In fact $\Pi_1(\mathbf{S}^1, 1)$ is isomorphic to $\mathbf{Z}$. Here is a strategy for proving this, and you should carry out the details. Recall that the covering space $(\mathbf{R}, \mathbf{S}^1, \pi)$, where $\pi(t) = e^{2\pi i t}$ has $\mathbf{Z}$ as its fiber over 1. If σ is any loop in $\mathbf{S}^1$ based at 1 we can lift it uniquely to a path γ starting at $0 \in \mathbf{R}$, and $n = \gamma(1) \in \mathbf{Z}$, Moreover we know that the homotopy class of γ with fixed endpoints depends only on $[\sigma]$, the homotopy class of σ with fixed endpoints, hence in particular n depends only on $[\sigma]$ and we define a map $\deg : \Pi_1(\mathbf{S}^1, 1) \to \mathbf{Z}$ by $\deg([\sigma]) = n$. If $k \in \mathbf{Z}$ then a path from 0 to k in $\mathbf{R}$ projects to a loop σ of degree n, so deg is surjective. If $\deg([\sigma]) = 0$, then γ is a loop in $\mathbf{R}$ based at 1, and since $\mathbf{R}$ is contractible, γ and hence σ is contractible with fixed endpoints to a constant loop. This will prove that deg is a monomorphism and hence an isomorphism of groups once it is shown that $\deg([\sigma_1] * [\sigma_2]) = \deg([\sigma_1] + \deg([\sigma_2])$. This is straightforward and it is left to you to carry out the details.

Lecture 29.
Applications of The Non-triviality of $\Pi_1(\mathbf{S}^1, 1)$

Note. When I gave these lectures at UC Irvine in the Fall of 2009, there were only twenty-eight lectures, but at the end of the final lecture I gave a brief preview of the following material, promising a more detailed treatment in the written notes. This "virtual lecture" is meant to fulfill that commitment.

29.1. Retracts and Deformation Retracts.

Definition. Let S be a subspace of a topological space X. A continuous map $\rho : X \to S$ is called a *retraction* of X on S if ρ restricted to S is the identity map of S.

29.2. The Brouwer Fixed-Point Theorem for $\mathbf{S}^2$.

29.2.1 The Covering Homotopy Theorem. *Let (E, B, π) be a covering space, $f : X \to B$ a continuous map of a space X into the base space, and $h : X \times I \to B$ a homotopy of f. If there is a lifting $F : X \to E$ of f, then there is a unique lifting $H : X \times I \to E$ of h to a homotopy of F. Moreover if the homotopy h is fixed on a subset A of X then its lifting H is also fixed on A.*

Proof. Note that if X is a point then this reduces to the Path Lifting Theorem, since a homotopy of a map of a point is the same thing as a path. In fact we can use the Path Lifting Theorem to prove the more general Covering Homotopy Theorem. For $x \in X$, let $t \mapsto H(x, t)$ be the unique path in E that starts at $F(x)$ and covers the path $t \mapsto h(x, t)$. The only thing left to prove is that this map H is continuous. Assume a sequence x_n converges to x in X. Then the subset Y of X consisting of the x_n and x is compact, so h is uniformly continuous on $Y \times I$, and hence the sequence of paths $t \mapsto h(x_n, t)$ converges uniformly to the path $t \mapsto h(x, t)$. But then by the Path Lifting Theorem, $t \mapsto H(x_n, t)$ converges uniformly to $t \mapsto H(x, t)$, proving that H is continuous. ∎

29.2.2 Corollary. *Let (E, B, π) be a covering space and let σ_0 and σ_1 are two paths in B with the same endpoints that are homotopic with fixed endpoints. If $\tilde{\sigma}_0$ and $\tilde{\sigma}_1$ are liftings of σ_0 and σ_1 respectively with the same initial point, then $\tilde{\sigma}_0$ and $\tilde{\sigma}_1$ are also homotopic with fixed endpoints, and in particular $\tilde{\sigma}_0(1) = \tilde{\sigma}_1(1)$.*

29.2.3 Corollary. *Let (E, B, π) be a covering space and let σ be a loop in B that is homotopic in B with fixed endpoints to a constant loop. Then any lift of σ is a loop in E that is homotopic in E with fixed endpoints to a constant loop.*

29.2.4 Corollary. *Let (E, B, π) be a covering space, $e_0 \in E$ and $b_0 = \pi(e_0) \in B$. Then $\Pi_1(\pi) : \Pi_1(E, e_0) \to \Pi_1((B, b_0)$, the induced map of fundamental groups, is injective with image the homotopy classes of loops in B based at b_0 that lift to loops based at e_0.*

▹ **Exercise 29–1.** The proofs of all the above corollaries are straightforward, but you should carry out the details to make sure that you understand all the new concepts.

29.2.5 Theorem. *If a pathwise connected space B has a non-trivial covering space (E, B, π) with a pathwise connected total space E, then $\Pi_1(B, b_0)$ is not the trivial group.*

Proof. Let γ be a path in E joining two distinct points of $\pi^{-1}(b_0)$. Then its projection $\sigma := \pi \circ \gamma \in \Lambda(B, b_0)$ is a loop in B whose lift to a path in E is not a loop in E, hence its homotopy class $[\sigma]$ is not in the image of $\Pi_1(\pi) : \Pi_1(E, e_0) \to \Pi_1(B, b_0)$ and in particular it is not the identity element of $\Pi_1(B, b_0)$. ∎

29.2.6 Corollary. $\Pi_1(\mathbf{S}^1, 1)$ *is non-trivial.*

▷ **Exercise 29–2.** In fact $\Pi_1(\mathbf{S}^1, 1)$ is isomorphic to $\mathbf{Z}$. Here is a strategy for proving this, and you should carry out the details. Recall that the covering space $(\mathbf{R}, \mathbf{S}^1, \pi)$, where $\pi(t) = e^{2\pi i t}$ has $\mathbf{Z}$ as its fiber over 1. If σ is any loop in $\mathbf{S}^1$ based at 1 we can lift it uniquely to a path γ starting at $0 \in \mathbf{R}$, and $n = \gamma(1) \in \mathbf{Z}$, Moreover we know that the homotopy class of γ with fixed endpoints depends only on $[\sigma]$, the homotopy class of σ with fixed endpoints, hence in particular n depends only on $[\sigma]$ and we define a map $\deg : \Pi_1(\mathbf{S}^1, 1) \to \mathbf{Z}$ by $\deg([\sigma]) = n$. If $k \in \mathbf{Z}$ then a path from 0 to k in $\mathbf{R}$ projects to a loop σ of degree n, so deg is surjective. If $\deg([\sigma]) = 0$, then γ is a loop in $\mathbf{R}$ based at 1, and since $\mathbf{R}$ is contractible, γ and hence σ is contractible with fixed endpoints to a constant loop. This will prove that deg is a monomorphism and hence an isomorphism of groups once it is shown that $\deg([\sigma_1] * [\sigma_2]) = \deg([\sigma_1] + \deg([\sigma_2])$. This is straightforward and it is left to you to carry out the details.